

[INSERT TITLE HERE]

JOHN CHRISTOPHER LENTFER KRISTIANSEN
202005794

MASTER THESIS IN MATHEMATICS
6. JANUARY 2024



SUPERVISOR: FABRICE BAUDOIN

INSTITUT FOR MATEMATIK
AARHUS UNIVERSITET

Contents

Contents	i
1 We gotta start somewhere	1
1.1 Iterated Function System	1
1.2 Construction the Vicsek set	2
1.3 Hausdorff Dimensions	3
1.4 Hausdorff, walk and spectral dimension of the Vicsek set	6
2 And now for something completely different	9
2.1 We're going to define it again	9
2.2 And back again	16
2.3 Discrete Fractional Gaussian Fields	16
2.4 Bounds of the fractional Laplacian	24
3 Fractional Gaussian Fields	27
3.1 Discrete fractional Gaussian fields	27
3.2 Fractional Gaussian Fields	28
4 Main Theorem	33
4.1 Preliminary lemmas	33
4.2 Main Theorem	35
4.3 Simulations	35
Bibliography	39

Chapter 1

We gotta start somewhere

In this thesis we will be studying the Vicsek set, so therefore we will in this section start by constructing the set and proving some basic properties of the Vicsek set. While it is possible to define the Vicsek set as a non-empty compact self-similar set, we will first define and prove some basic properties of self-similar sets.

1.1 Iterated Function System

The iterated functions systems (IFS) gives us a way to work with self-similarly sets. Another reason for constructing the IFS is that we are able to prove that the Vicsek set actually exists.

Definition 1.1: A finite family of functions $\{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_m\}$ with $m \geq 2$ is called an *iterated function system* (or IFS) if exists $c_i \in (0, 1)$ such that all $\mathcal{S}_i$ are *contractions*. Which is exactly

$$|\mathcal{S}_i(x) - \mathcal{S}_i(y)| \leq c_i |x - y|.$$

Further we call a non-empty compact subset K of X an *invariant set* for the IFS if

$$K = \bigcup_{i=1}^m \mathcal{S}_i(K) \tag{1.1}$$

The next question will then be if there for every IFS exists an invariant set for the IFS? We shall actually prove that this is indeed true, but before we begin the proof that a unique invariant set exists we will need to define the Hausdorff metric.

Definition 1.2: Define $A_\delta = \{x \in \mathcal{A} : \exists a \in A, |x - a| < \delta\}$ The *Hausdorff metric* is given by

$$d_H(X, Y) = \inf \{\delta : X \subset Y_\delta \text{ and } Y \subset X_\delta\}$$

The proof that d_H metric can be found in [Hen99].

Theorem 1.3: Consider the IFS given by $\{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_m\}$, then there is a unique invariante set K for the IFS.

Moreover, if we define a transformation S , on the class $\mathcal{A}$ of non-empty compact sets, by

$$S(E) = \bigcup_{i=1}^m \mathcal{S}_i(E), \quad \text{for } E \in \mathcal{A}$$

and write S^k to be the k 'th iterate of S , so $S^0 = E$ and $S^k(E) = S(S^{k-1}(E))$ for $k \geq 1$. Then

$$F = \bigcap_{i=0}^{\infty} S^i(E)$$

for every set $E \in \mathcal{A}$ such that $\mathcal{S}_i(E) \subset E$ for all i .

Proof. Let E be any set in $\mathcal{A}$ such that $S_i(E) \subset E$ for all i . By induction we find that $S^k(E) \subset S^{k-1}(E)$. Since $S^0 = E \supset S(E) = S^1(E)$, then $S^{k+1}(E) = S(S^k(E)) \subset S(S^{k-1}(E)) = S^k(E)$ by the induction assumption that $S^k \subset S^{k-1}$. So $S^k(E)$ is a decreasing sequence of non-empty compact sets, in which case it must have a non-empty compact intersection $K = \bigcap_{k \geq 0} S^k(E)$. Since $S^k(E)$ is decreasing, then $S(K) = K$ fulfilling (1.1).

Now moving on to uniqueness, We will need the following inequality

$$d_H(S(A), S(B)) = d_H\left(\bigcup_{i=1}^m S_i(A), \bigcup_{i=1}^m S_i(B)\right) \leq \max_{1 \leq i \leq m} d_H(S_i(A), S_i(B)) \leq \max_{1 \leq i \leq m} c_i \cdot d_H(A, B). \quad (1.2)$$

So assume A and B are both invariant sets, then $S(A) = A$ and $S(B) = B$, but that gives us that

$$d_H(A, B) = d_H(S_i(A), S_i(B)) \leq \max_{1 \leq i \leq m} c_i \cdot d_H(A, B)$$

and since $0 < \max_{1 \leq i \leq m} c_i < 1$, then $d_H(A, B) = 0$ so $A = B$ and it is unique.

Lastly we will prove (1.2). The first inequality comes by choosing $\delta = \max_{1 \leq i \leq m} d_H(S_i(A), S_i(B))$ since that means that $S_i(B) \subset (S_i(A))_\delta$ for all $0 \leq i \leq m$, thus it will definitely hold for $\bigcup_{i=1}^m S_i(B) \subset (\bigcup_{i=1}^m S_i(A))_\delta$ and vice versa. The second inequality comes from letting $\delta = d(A, B)$ then for every $a \in A$ we can choose some $b \in B$ such that $|a - b| \leq \delta$ in which case $|S_i(a) - S_i(b)| \leq c_i \delta$ and therefore $S_i(A) \subset (S_i(B))_{c_i \delta}$. $\square$

In the end we will mention a few consequences of of theorem 1.3. The theorem gives us a way to create a (not necessarily unique) sequence $(i_1, i_2, \dots)$ such that each $x \in F$ we can write

$$\{x\} = \{x_{i_1, i_2, \dots}\} = \bigcap_{k \geq 0} K_{i_1, \dots, i_k}.$$

This is possible since S^k is decreasing as k increases. This further lets us describe K as

$$K = \bigcup_{\mathcal{I}} \{x_{i_1, i_2, \dots}\}. \quad (1.3)$$

1.2 Construction the Vicsek set

Let $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ be the set of vertices in a plus with a middle point in $\mathbb{R}^2$. Then define

$$S_i(z) = \frac{1}{3}(z - p_i) + p_i, \quad \text{for } i = 1, 2, \dots, 5.$$

$\{S_1, \dots, S_5\}$ is indeed a IFS, since for every $x, y \in \mathbb{R}$

$$|S_i(x) - S_i(y)| = \left| \frac{1}{3}(x - p_i) + p_i - \left(\frac{1}{3}(y - p_i) + p_i \right) \right| = \frac{1}{3}|x - y|.$$

Thus by theorem 1.3 we define the Vicsek set K is as the unique non-empty compact subset in $\mathbb{R}^2$ such that

$$K = \bigcup_{i=1}^5 S_i(K).$$

We note that V_0 is the boundary of K .

Since we are going studying the convergence of discretely defined operators to smooth operators we can define a discrete Vicsek set in the follow way. We start by define a sequence of sets $(V_m)_{m \geq 0}$ inductively:

$$V_{m+1} = \bigcup_{i=1}^5 S_i(V_m)$$

this gives us a sequence of Vicsek set graphs $(G_m)_{m \geq 0}$ whose vertices are $(V_m)_{m \geq 0}$. Further we will use the notation of $V_* = \bigcup_{m \geq 0} V_m$ and for any $p \in V_m$ we denote the collection of neighbours of p in G_m by $V_{m,p}$. Lastly note that $N_m = \#V_m = 5^m \cdot 4 + 1$.

1.3 Hausdorff Dimensions

What is hausdorff dimension? We first have to look at Hausdorff measure.

We say the countable collection of set $\{U_i\}_{i \geq 1}$ is a δ -cover of K if $K \subset \bigcup_{i \geq 1} U_i$ and $\text{diam } U_i \leq \delta$. Here $\text{diam}(A) := \sup_{x,y \in A} d(x,y)$ where $d(\cdot, \cdot)$ is a metric.

Definition 1.4: Let X be a metric space. If $K \subset X$ and $d \in [0, \infty)$ we define

$$H_\delta^d(K) = \inf \left\{ \sum_{i=1}^{\infty} \text{diam}(U_i)^d : \bigcup_{i=1}^{\infty} U_i \supseteq K, \text{diam}(U_i) < \delta \right\}.$$

Then the d -dimensional *Hausdorff measure* is given by

$$\mathcal{H}^d(K) = \lim_{\delta \rightarrow 0} H_\delta^d(K)$$

The Hausdorff measure defined above, is not a measure in the normal measure theory sense but rather it is an outer measure. Though it is possible to prove that the Hausdorff measure restricted to Borel-set is a measure.

Moving swiftly along, we now want to look at the behaviour of $\mathcal{H}^s$ as s changes. So let $U_{i \geq 1}$ is a δ -cover of K then we have for $0 \leq s < t$ that

$$\sum_{i \geq 1} \text{diam}(U_i)^t = \sum_{i \geq 1} \text{diam}(U_i)^{t-s} \text{diam}(U_i)^s \leq \delta^{t-s} \sum_{i \geq 1} \text{diam}(U_i)^s.$$

Taking infima on both sides we get that $H_\delta^t(K) \leq \delta^{t-s} H_\delta^s(K)$. If we let $\delta \rightarrow 0$ we see that if $\mathcal{H}^s(K) < \infty$ then $\mathcal{H}^t(K) = 0$ since $\delta^{t-s} \rightarrow 0$. Likewise if $\mathcal{H}^t > 0$ then $\mathcal{H}^s = \infty$. This tells us that if there is a point, say κ , such $\mathcal{H}^\kappa$ is both positive and finite, then $\mathcal{H}^s$ will be 0 for all $s > \kappa$ and ∞ for all $s < \kappa$. That such value is called the *Hausdorff Dimension* of K . Formally written:

Definition 1.5: The *Hausdorff dimension*, denoted $d_h(K)$ is given by

$$d_h(K) = \inf \{s \geq 0 : \mathcal{H}^s(K) = 0\} = \sup \{s \geq 0 : \mathcal{H}^s(K) = \infty\}.$$

When there is no confusion with regard to what set we are working with, we denote the Hausdorff dimension by d_h .

Now it is quite difficult to determine the Hausdorff dimension from the definition. Luckily we can with a few assumption find the Hausdorff dimension quite easily. For that theorem we need an important lemma.

Lemma 1.6: Let μ be a measure such that $\mu(F) \in (0, \infty)$ and suppose there are numbers $c, \epsilon > 0$ such that for some $s \geq 0$

$$\mu(U) \leq c \text{diam}(U)^s$$

for all sets U with $\text{diam}(U) \leq \epsilon$. Then $\mathcal{H}^s(F) \geq \frac{\mu(F)}{c}$ and $s \leq d_h$

Proof. If $\{U_i\}_{i \in I}$ is any cover of F then

$$0 < \mu(F) \leq \mu \left(\bigcup_{i \in I} U_i \right) \leq \sum_{i \in I} \mu(U_i) \leq c \sum_{i \in I} \text{diam}(U_i)^s$$

by definition of measures and properties of μ . Taking infimum we get $\mathcal{H}_\delta^s \geq \frac{\mu(F)}{c}$ and therefore $\mathcal{H}^s \geq \frac{\mu(F)}{c} > 0$ so by definition 1.5 $s \leq d_h$. $\square$

Lemma 1.7: Let $\{V_i\}$ be a collection of disjoint open subset of $\mathbb{R}^n$ such that each V_i contains a ball of radius $a_1 r$ and is contained in a ball of radius $a_2 r$. Then any ball B of radius R intersects at most $(1 + 2a_2)^n a_1^n$ of the closures $\bar{V}_i$

Proof. By assumption we know that $\text{diam}(V_i) \leq 2a_2r$ for all V_i . So if some $\bar{V}_i$ intersect the ball with radius r , then $\bar{V}_i$ must be contained in the ball with radius $(2a_2 + 1)r$ concentric with the ball with radius r . Now let q denote the number of closures $\bar{V}_i$ that are intersecting with the ball with radius r . Then there exists q disjoint balls with radius a_1r that are contained in the ball with radius $(2a_2 + 1)r$, since by assumption each $\bar{V}_i$ contains a ball of radius a_1r . It then follows that $q(a_1r)^n \leq (2a_2 + 1)^n r^n$ which gives us the bound for q . $\square$

We first define the open set condition

Definition 1.8: We say that a IFS $\mathcal{S}_{i=1 \leq i \leq m}$ satisfy the *open set condition* if there exists a non-empty bounded open set V such that

$$V \supset \bigcup_{i=1}^m \mathcal{S}_i(V)$$

and each $\mathcal{S}_i(V)$ is disjoint.

Theorem 1.9: Suppose definition 1.8 holds for the similarities $\mathcal{S}_i$ on $\mathbb{R}^n$ with ratios $0 < c_i < 1$ for $1 \leq i \leq m$. If there for K holds

$$K = \bigcup_{i=1}^m \mathcal{S}_i(K) \tag{1.4}$$

then $d_h = s$ where s is given by

$$\sum_{i=1}^m c_i^s = 1. \tag{1.5}$$

Moreover, for this value of s , $0 < \mathcal{H}^s(K) < \infty$

Proof. Let $\mathcal{I}_k$ denote the set of all k -term sequences $(i_1, \dots, i_k)$ with $1 \leq i_j \leq m$. For any set A and any sequence $(i_1, \dots, i_k) \in \mathcal{I}_k$ we denote $A_{i_1, \dots, i_k} = \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(A)$.

We will start by finding an upper bound for the Hausdorff measure. We do this by finding an appropriate δ -cover for K . It follows that by repeated use of (1.4) that

$$K = \bigcup_{i=1}^m \mathcal{S}_i(K) = \bigcup_{i=1}^m \mathcal{S}_i \left(\bigcup_{j=1}^m \mathcal{S}_j(K) \right) = \bigcup_{i,j=1}^m \mathcal{S}_i \circ \mathcal{S}_j(K) = \dots = \bigcup_{\mathcal{I}_k} \mathcal{S}_{i_1, \dots, i_k}(K) = \bigcup_{\mathcal{I}_k} K_{i_1, \dots, i_k}.$$

Now noting that $\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}$ has the ratios $c_{i_1} \dots c_{i_k}$, we see

$$\sum_{\mathcal{I}_k} \text{diam}(K_{i_1, \dots, i_k})^s = \sum_{\mathcal{I}_k} (c_{i_1} \dots c_{i_k})^s \text{diam}(K)^s = \left(\sum_{i_1} c_{i_1}^s \right) \dots \left(\sum_{i_k} c_{i_k}^s \right) \text{diam}(K)^s = \text{diam}(K)^s$$

where the last equality comes from (1.5). Now for any $\delta > 0$ we can choose k so

$\text{diam}(K_{i_1, \dots, i_k}) (\max_{1 \leq i \leq m} c_i)^k \text{diam}(K) \leq \delta$ since $\max_{1 \leq i \leq m} c_i \in (0, 1)$. Thus $\{K_{i_1, \dots, i_k}\}_{\mathcal{I}_k}$ is a δ -cover for K thus $\mathcal{H}_\delta^s(K) \leq \text{diam}(K)^s$ in which case $\mathcal{H}^s(K) \leq \text{diam}(K)^s$ and we have an upper bound.

Next we need a lower bound for the Hausdorff dimension, which is substantially more difficult.

Let $\mathcal{I} = \{(i_1, i_2, \dots) : 1 \leq i_j \leq m\}$ be the set of all infinite sequences and let $\mathcal{I}_{i_1, \dots, i_k} = \{(i_1, \dots, i_k, q_{k+1}, \dots) : 1 \leq q_j \leq m\}$, so the first initial k "terms" are chosen and then the rest are the remaining permutations.

Now we create an outer measure μ on subset of $\mathcal{I}$ defined by $\mu(\mathcal{I}_{i_1, \dots, i_k}) = (c_{i_1} \dots c_{i_k})^s$. From this outer measure μ we construct an outer measure on K named $\tilde{\mu}$ defined in the following way. For some subset $A \subseteq K$ -

$$\tilde{\mu}(A) = \mu \left(\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1, i_2, \dots} \in A \cap K\} \right)$$

Proofs that μ and $\tilde{\mu}$ are indeed outer measures can be found in [Haz22, Section 7].

Now we want to show that $\tilde{\mu}$ fulfills the properties of lemma 1.6.

Let V be the open set that satisfies definition 1.8. Since $\bar{V} \supset \mathcal{S}(\bar{V}) = \bigcup_{i=1}^m \mathcal{S}_i(\bar{V})$, then by theorem 1.3 the decreasing sequence $\mathcal{S}^k(\bar{V})$ converges to K . In particular we have $\bar{V} \supset \mathcal{S}(\bar{V}) \supseteq \bigcup_{i=1}^m \mathcal{S}_i(\bar{V}) = F$ and $\bar{V}_{i_1, \dots, i_k} = \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(\bar{V}) \supseteq \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(F) = F_{i_1, \dots, i_k}$ for each finite sequence $(i_1, \dots, i_k)$.

Now let B be any ball $r < 1$. We estimate $\tilde{\mu}(B)$ by considering sets $V_{i_1, \dots, i_k}$ with diameters comparable with that of B with closures intersecting $F \cap B$. Further we define $\mathcal{Q}$ as the set of sequences $(i_1, \dots, i_k) \in \mathcal{I}$ such that i_k is the smallest k for which the following holds

$$\left(\min_{1 \leq i \leq m} c_i \right) r \leq c_{i_1} \dots c_{i_k} \leq r.$$

Note that no sequence in $\mathcal{Q}$ extends to another sequence. Since if two sequences are the same, but one is longer than the other the shorter sequence would be the one in $\mathcal{Q}$. Now by definition 1.8 $V_1, \dots, V_m$ are disjoint and so are $V_{i_1, \dots, i_k, 1}, \dots, V_{i_1, \dots, i_k, m}$ for each $(i_1, \dots, i_k)$. In particular we have that $\{V_{i_1, \dots, i_k} : (i_1, \dots, i_k) \in \mathcal{Q}\}$ is a collection of disjoint sets. We further note that by (1.3) we have

$$K = \bigcup_{\mathcal{I}} \{x_{i_1, i_2, \dots}\} \subseteq \bigcup_{\mathcal{Q}} K_{i_1, \dots, i_k} \subseteq \bigcup_{\mathcal{Q}} \bar{V}_{i_1, \dots, i_k}. \quad (1.6)$$

Next we choose a_1 and a_2 so that V contains a ball of radius a_1 and is contained in a ball of radius a_2 . Then for all $(i_1, \dots, i_k) \in \mathcal{Q}$ the set $V_{i_1, \dots, i_k}$ contains a ball of radius $c_{i_1} \dots c_{i_k} a_1$ and therefore one of radius $(\min_{1 \leq i \leq m} c_i) a_1 r$. Further it is contained in a ball of radius $c_{i_1} \dots c_{i_k} a_2$ and therefore one of radius $a_2 r$.

Now let $\mathcal{Q}_B$ be the set of all sequences $(i_1, \dots, i_k) \in \mathcal{Q}$ such that $\bar{V}_{i_1, \dots, i_k}$ intersects B . By lemma 1.7 there are at most

$$q = (1 + 2b)^n \left(\min_{1 \leq i \leq m} c_i \right)^{-n} a^{-n}$$

sequences in $\mathcal{Q}_B$. Thus we find that if

$$x_{i_1, \dots, i_k} \in F \cap B \subseteq \bigcup_{\mathcal{Q}_B} \bar{V}_{i_1, \dots, i_k}, \quad \text{by (1.6),}$$

then there must be some k such that $(i_1, \dots, i_k) \in \mathcal{Q}_B$ and therefore $(i_1, i_2, \dots) \in \bigcup_{\mathcal{Q}_B} I_{i_1, \dots, i_k}$. We can thus conclude that

$$\tilde{\mu}(B) = \mu \left(\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1, i_2, \dots} \in F \cap B\} \right) \leq \mu \left(\bigcup_{\mathcal{Q}_B} I_{i_1, \dots, i_k} \right).$$

It then follows that

$$\tilde{\mu}(B) \leq \sum_{\mathcal{Q}_B} \mu(I_{i_1, \dots, i_k}) = \sum_{\mathcal{Q}_B} (c_{i_1} \dots c_{i_k})^s \leq \sum_{\mathcal{Q}_B} r^s \leq q r^s.$$

Since any set U is contained in a ball of radius $\text{diam}(U)$, we have that $\tilde{\mu} \leq q \text{diam}(U)^s$. Thus by lemma 1.6 we have that $\mathcal{H}^s(F) \geq \frac{\tilde{\mu}(F)}{q} = q^{-1} > 0$, thus $0 < q^{-1} \leq \mathcal{H}^s(F) \leq \text{diam}(F)^s < \infty$ thus by definition 1.5 we have that $s = d_h$ $\square$

Spectral and Walk dimension

Further in this thesis we will bound certain random fields and functions by a constant we call the spectral dimension. It depends on two other factors, namely the Hausdorff dimension, which we have just found, and of the walk dimension. Properly explaining the walk dimension is outside the scope of this thesis, but we will give some intuition as to what it is.

Let B_t be a Brownian motion on K , and $B(x, r)$ be the ball centered at $x \in K$ with radius r . then the *walk dimension* is the constant $d_w > 0$ such that

$$c_1 r^{d_w} \leq \mathbb{E} [\tau(x, r)] \leq c_2 r^{d_w}$$

where $\tau(x, r) = \inf \{t \geq 0 : B_t \notin B(x, r)\}$. The intuition is that it is the average escape rate, meaning how fast does the Brownian motion escape the ball. This explanation for the walk dimension is a rather simplified and condensed one, for further reference see [Mar98, Chapter 3].

This then leads us to being able to define the spectral dimension as following:

Definition 1.10: Let d_h be the Hausdorff dimension and d_w the walk dimension then the spectral dimension is

$$d_s = \frac{2d_h}{d_w}$$

1.4 Hausdorff, walk and spectral dimension of the Vicsek set

Lastly we need calculate the Hausdorff, walk and spectral dimension for the Vicsek set. Recalling that the Vicsek set is the unique set K such, for the iterative functions system $\{\mathcal{S}_1, \dots, \mathcal{S}_5\}$ with $\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i$ where $p_i \in V_0$, it holds that

$$K = \bigcup_{i=1}^5 \mathcal{S}_i(K).$$

We want to use theorem 1.9 to find the Hausdorff dimension so we need to check that $\mathcal{S}_{i1 \leq i \leq 5}$ satisfies definition 1.8.

So if we choose the open set V to be the open box where each corner is a point in V_0 (see section 1.4), then by continuity of $\mathcal{S}_i$ we have that $\mathcal{S}_i(V)$ is an open and disjoint. The disjointness of $\mathcal{S}_i$ is that each $\mathcal{S}_i$ "retracts" the whole set towards the point p_i and we get that the sets are disjoint. One might worry about the point where the open sets meet, but since the points in V_0 (excluding the middle point) are not included in the initial open box we have that they don't intersect.

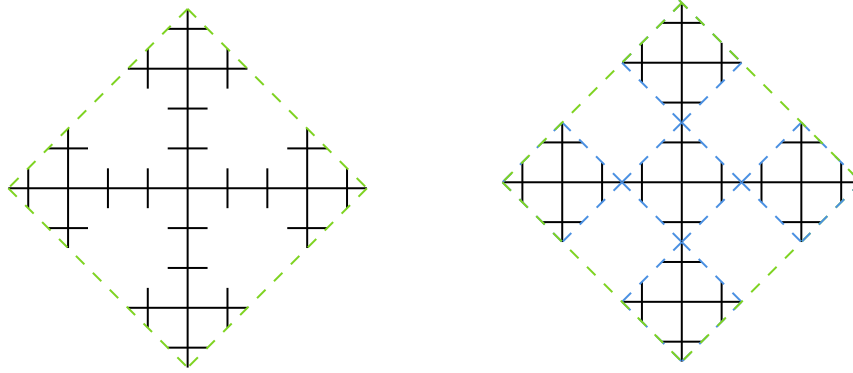


Diagram 1.1: The opens sets V (green) and $\mathcal{S}_i(V)$ (blue)

Thus we can conclude from theorem 1.9 that

$$\sum_{i=1}^5 \left(\frac{1}{3}\right)^{d_h} = 1 \Rightarrow \left(\frac{1}{3}\right)^{d_h} = \frac{1}{5} \Rightarrow d_h = \frac{\log 5}{\log 3}.$$

Thus the Hausdorff dimension for the Vicsket set is $\log 5 / \log 3$.

As stated earlier we will not be calculating the walk dimension, but instead just state it. The walk dimension for the Vicsek set is $d_w = d_h + 1$. Thus we can make the following definition

Definition 1.11: The following constants hold for the Vicsek set K

$$d_h = \frac{\log 5}{\log 3}, \quad d_w = d_h + 1, \quad \text{and} \quad d_s = \frac{d_h}{2d_w} = \frac{\log 5}{2 \log 15}$$

Having just defined the Hausdorff dimension we will immediately use it to construct the following measure.

Definition 1.12: The normalized Hausdorff measure on K which is given by the borel measure μ on K such for any $i_1, \dots, i_n \in \{1, \dots, 5\}$

$$\mu(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) = 5^{-n}.$$

We have the following property of the measure μ

Proposition 1.13: This measure μ is d_h -Ahlfors regular, i.e there exists $c_1, c_2 > 0$ such there for every $x \in K$ and $r \in [0, \text{diam}(K)]$

$$c_1 r^{d_h} \leq \mu(B(x, r)) \leq c_2 r^{d_h}.$$

Proof. If $r = 0$ the proof is obvious. So let $r \in (0, 1)$, and choose n so $R \cdot 3^{-(n+1)} \leq r \leq R \cdot 3^n$ where $R = \text{diam}(K)$. Note that for a sequence we have

$$\text{diam}(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) = 2 \cdot 3^{-n} \cdot R.$$

We can therefore construct a sequence $i_1, \dots, i_n$ and $i_1, \dots, i_{n+1}$ such that $B(x, r) \supset \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{n+1}}$ and $\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n} \cap B(x, r) \neq \emptyset$. Since $B(x, r)$ intersects with at most 5 of these sequences we find an upper bound to be.

$$\mu(B(x, r)) \leq 5 \cdot \mu(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) = 5 \cdot 5^{-n} = 5^2 \cdot 5^{-(n+1)} \leq 5 \cdot \left(R \cdot 3^{-(n+1)}\right)^{d_h} \leq 5 \cdot r^{d_h}.$$

thus we have an upper bound. This bound also works for $R < 1$ since then we just omit the constant from the calculations and have $3^{-(n+1)} \leq R^{-1}r$ and the bound becomes $5 \cdot (R^{-1}r)^{d_h}$. Likewise for the lower bound we have that there exists a sequence $i_1, \dots, i_{n+1}$ such that $B(x, r) \supset \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}$ in which case

$$\mu(B(x, r)) \geq \mu(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{n+1}}(K)) = 5^{-(n+1)} \geq R \cdot 5^{-(n+1)} \geq 5 \cdot \left(R^{-1}r\right)^{d_h}.$$

Again this bound also holds for $R > 1$ since we again omit R in the calculations have $R^{-1}r \leq 3^n$ and the bound $5 \cdot (R^{-1}r)^{d_h}$.

Now assume $\text{diam}(K) \geq r \geq 1$ the upper bound is simply $\mu(B(x, r)) \leq \mu(K) = 1 \leq r$. For the lower bound consider that there exists $m \in \mathbb{N}$ so $3^{m-1} \leq r \leq 3^m$, then

$$\mu(B(x, r)) \geq \mu(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{m-1}}(K)) = 5^{-(m-1)} = 5 \cdot (3^{-m})^{d_h} \geq (r3^{-2m})^{d_h}$$

as desired. $\square$

Next we have that we define a sequence of discrete measures

Definition 1.14:

$$\mu_m = \frac{1}{a_m} \sum_{p \in V_m} \delta_p \quad (1.7)$$

where $a_m = 5^m \cdot 4 + 1$.

We have that $\mu_m \rightarrow \mu$ as $m \rightarrow \infty$, this follows from the fact that $\mathcal{S}_i(K)$ "shrinks" the fractal down to $1/5$ the size around the point p_i . In which case for any sequence $i_1, \dots, i_n$ we have that

$$\begin{aligned} \lim_{m \rightarrow \infty} \mu_m(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) &= \lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} \delta_p(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) \\ &= \lim_{m \rightarrow \infty} \frac{1}{a_m} \cdot 5^{-n} \cdot (5^m \cdot 4 + 1) = 5^{-n} = \mu(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) \end{aligned}$$

Which leads us to the following corollary. **Wrote something wrong here maybe? wrt. to converge weakly vs converge in weak topology**

Corollary 1.15: The sequence of measure $(\mu_m)_{m \geq 0}$ defined in (1.7) converges weakly to the normalized Hausdorff measure μ on the Vicsek set K . That is

$$\lim_{m \rightarrow \infty} \int_K f \, d\mu_m = \int_K f \, d\mu$$

Chapter 2

And now for something completely different

In future section the Gaussian random fields will be defined using the eigenvalues and eigenfunctions of the Laplacian. We will therefore start by introducing first the discrete Laplacian, then the Kigami Laplacian and will culminate in the Dirichlet Laplacian.

We will now consider the discrete Laplacian on the fractal.

Definition 2.1: Let $\ell(V_m)$ be the set of functions of $f : V_m \rightarrow \mathbb{R}$. Then for any $f \in \ell(V_m)$ the *discrete Laplacian* on V_m is defined as

$$\Delta_m f(p) = 3^m \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \setminus V_0$$

Remark 2.2: One is able to describe the discrete Laplacian as a matrix. First let $p_1, \dots, p_{N_m}$ be the vertices in V_m then the *Laplacian matrix* L_m is constructed in the following way:

$$L_{i,j} = \begin{cases} \#V_{m,p_i}, & \text{if } i = j \\ -1, & \text{if } i \neq j \text{ and } p_j \in V_{m,p_i} \\ 0, & \text{otherwise.} \end{cases}$$

Further $\#V_{m,p_i}$ is also called the *degree* of p_i

The discrete Laplacian will be used to construct the Dirichlet Laplacian which is the one we will be working with in the rest of this thesis. The construction require the construction of the Kigami Laplacian, which we will define as a pointwise limit of the discrete Laplacian.

2.1 We're going to define it again

In this section we will follow arguments given by [Str06]. Much of the proof are going to rely on the standard graph Dirichlet form. Thus we are first going to define and prove some properties of this graph Dirichlet form.

Definition 2.3: Given a finite connected graph G and it vertices V_m the for functions $u, v : V_m \rightarrow \mathbb{R}$ we define the *standard graph Dirichlet form* to be

$$E_m(u, v) = \sum_{x \sim y} (u(x) - u(y)) (v(x) - v(y)).$$

Notation: We will write $E_m(u) = E_m(u, u)$.

Since we in the end want to work on the whole of the Vicsek set K . We will want to create an extension of a function $u : V_m \rightarrow \mathbb{R}$, such that it is defined on V_{m+n} . This is done in the following way

Definition 2.4: Let K be a graph and $u : V_m \rightarrow \mathbb{R}$, then an *extension* of u is a function $u' : V_{m+n} \rightarrow \mathbb{R}$ where $u' |_{V_m} = u$.

It is obvious that $E_{m+n}(u') \geq E_m(u)$. Further one quickly notices that there must be a function that minimizes the graph Dirichlet form of a given graph. We denote such a function with $\tilde{u}$ and call it a *harmonic extension*.

Definition 2.5: Let $\tilde{u} : V_{m+n} \rightarrow \mathbb{R}$ with $|_{V_m} = u$ is called a *harmonic extension* if $E_{m+n}(\tilde{u}) \leq E_{m+n}(u')$ for all $u' : V_{m+n} \rightarrow \mathbb{R}$ with $u' |_{V_m} = u$.

In some cases we have that $E_{m+n}(\tilde{u}) = r^n E_m(u)$ for some $r \in (0, 1)$. But instead of normalizing the values on E_m we will normalize E_{m+n} . Thus getting

$$r^{-n} E_{m+n}(\tilde{u}) = E_m(u), \quad \text{and} \quad r^{-n} E_{m+n}(u') \geq E_m(u)$$

for every extension u' of u . The constant r is called the *renormalization constant*. From the equality above we can define the following

Definition 2.6: The *renormalized graph energies* are defined as

$$\mathcal{E}_m(u) = r^{-m} E_m(u)$$

Note that the sequence $\{\mathcal{E}_m(u)\}$ is a non-decreasing sequence.

Based on this definition we see that for a harmonic extension $\tilde{u}$ of u we have that

$$\mathcal{E}_{m+1}(\tilde{u}) = r^{-(m+1)} E_{m+1}(\tilde{u}) = r^{-m} E_m(u) = \mathcal{E}_m(u).$$

Extending this to the bilinear form we have for harmonic extension $\tilde{u}, \tilde{v}$ of u, v that

$$\mathcal{E}_{m+1}(\tilde{u}, \tilde{v}) = \mathcal{E}_m(u, v).$$

Actually one even has that for the bilinear form then if $\tilde{u}$ is the harmonic extension of u and v' is any extension of v then

$$\mathcal{E}_{m+1}(\tilde{u}, v') = \mathcal{E}_m(u, v). \tag{2.1}$$

See [Str06] for further details.

Now that we have 2.6, we can define what it means to be a harmonic function.

Definition 2.7: u is a *harmonic function* if u minimizes $\mathcal{E}_m(u)$ for all m , given boundary values on V_0 .

The harmonic functions will be used extensively throughout this section. We start by finding the renormalization constant for the Vicsek set.

Renormalization constant on the Vicsek set

In the previous section we have talked about a harmonic function, but without every constructing one. We will therefore in this next section show how to create an harmonic function for the Vicsek set for any start boundary values.

Given $V_0 = \{q_0, q_1, q_2, q_3, q_4\}$ and $u : V_0 \rightarrow \mathbb{R}$ then we have that

$$E_0(u) = (u(q_0) - u(q_1))^2 + (u(q_0) - u(q_2))^2 + (u(q_0) - u(q_3))^2 + (u(q_0) - u(q_4))^2.$$

We are going to exploit symmetry of the Vicsek set. Since all four "limbs" of the Vicsek set are identical we will only look at one of the limbs since the other limbs will be the same computations. So if we want to minimize $E_1(u')$, we see for the limb between q_0 and q_1 that

$$\begin{aligned} E_1(u') &= (u'(q_1) - u'(p_0))^2 + (u'(p_0) - u'(p_2))^2 \\ &\quad + (u'(p_0) - u'(p_3))^2 + (u'(p_0) - u'(p_4))^2 + (u'(p_3) - u'(q_0))^2. \end{aligned}$$

We are then going to optimize with regards to $u'(p_i)$ for $i = 0, 1, 2, 4$. Taking the derivative of the right side and solving for 0 we get the following

$$\begin{aligned} 4u'(p_0) &= u'(q_1) + u'(p_2) + u'(p_3) + u'(p_4) \\ 2u'(p_3) &= u'(q_0) + u'(p_0) \\ u'(p_2) &= u'(p_0) \\ u'(p_4) &= u'(p_0). \end{aligned}$$

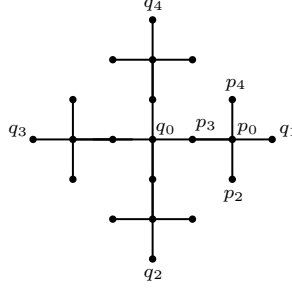


Diagram 2.1: Points on the Vicsek set

Solving these equations we get

$$\begin{aligned} u'(p_0) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_2) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_4) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_3) &= \frac{2}{3}u'(q_0) + \frac{1}{3}u'(q_1). \end{aligned}$$

So what we have found is that to minimize the value in the points of the Vicsek set, then the value in the next point is going to be $2/3$ the value from the closest point and $1/3$ the value from the next closest point.

If one calculates the standard Dirichlet form then you will get

$$E_1(u') = \frac{1}{3} \sum_{i=1}^4 (u'(q_i) - u'(q_0))^2 = \frac{1}{3} E_0(u').$$

Since the Vicsek set is a self-similar set, we have that going from V_1 to V_2 is the same as going from V_m to V_{m+1} . Thus we can just extend the above calculations to doing it m times, in which case

$$E_{m+1}(u') = \frac{1}{3} E_m(u')$$

Thus we get our renormalization constant for the vicsek set to be $r = 3$. And altogether we find that from definition 2.6 we have that

$$\mathcal{E}_m(u) = 3^{-m} E_m(u).$$

With the renormalization constant found we will return to finding the Kigmai Laplacian. Recall from definition 2.6 that we have defined a sequence $\mathcal{E}_m$ for functions on V^* . It will therefore make sense to define the following limit

Definition 2.8: Let $u : V^* \rightarrow \mathbb{R}$ then define the limit

$$\mathcal{E}(u, u) = \lim_{m \rightarrow \infty} \mathcal{E}_m(u, u)$$

where we set

$$\text{dom } \mathcal{E} = \{u \in \ell(V_*) : \mathcal{E}(u) < \infty\}.$$

and denote $\text{dom}_0 \mathcal{E}$ with the functions in $\text{dom } \mathcal{E}$ that vanish V_0 .

This definition only involves functions defined on V^* , but we find that functions in $\text{dom } \mathcal{E}$ are uniformly continuous and therefore has an unique extension to K . We even have that $\text{dom } \mathcal{E}$ is dense in $C(K)$.

Proposition 2.9:

- (i) Let $u \in \text{dom } \mathcal{E}$ then u is uniformly continuous on V^* .
- (ii) $\text{dom } \mathcal{E}$ is dense in $C(K)$.

Proof. We start with (item i). Let $u \in \text{dom } \mathcal{E}$ then $\mathcal{E}_m(u) \leq \mathcal{E}(u)$ since $\mathcal{E}_m$ is non-decreasing in m . So for $x, y \in V_m$ if $y \in V_{m,x}$ we have that

$$r^{-m}(u(x) - u(y))^2 \leq \mathcal{E}_m(u) \leq \mathcal{E}(u) \quad (2.2)$$

since $r^{-m}(u(x) - u(y))^2$ is a term in $\mathcal{E}_m(u)$. Thus

$$|u(x) - u(y)| \leq r^{\frac{m}{2}} \mathcal{E}(u)^{\frac{1}{2}}.$$

Now consider a chain of points $x_m, x_{m+1}, \dots, x_{m+k}$ where $x_{m+j} \in V_{m+j}$ and $x_{m+j} \in V_{m+j+1, x_{m+j+1}}$. Then we have

$$|u(x_m) - u(x_{m+k})| \leq r^{\frac{m}{2}} \left(1 + r^{\frac{1}{2}} + \dots + r^{\frac{k}{2}}\right) \mathcal{E}(u)^{\frac{1}{2}} \leq \frac{r^m}{1 - r^{\frac{1}{2}}} \mathcal{E}(u)^{\frac{1}{2}}$$

by using (2.2) and the triangle inequality multiple times. From the geometry of K we see that $x, y \in V^*$ belong to the same or adjacent m -cells then we can connect x to y by at most 2 such chains **What does he mean?**, so

$$|u(x) - u(y)| \leq \frac{2r^{m/2}}{1 - r^{\frac{1}{2}}} \mathcal{E}(u)^{\frac{1}{2}}$$

and we have a bound for all $x, y \in V^*$.

For proof of item ii we refer to [Str06]. □

We are now ready to define the Kigami Laplacian. We will denote this with Δ_μ and we will notice that it depends on a measure μ , which will be the one defined in (1.12).

Definition 2.10: Let $u \in \text{dom } \mathcal{E}$ and let f be continuous, then $u \in \text{dom } \Delta_\mu$ and $\Delta_\mu u = f$ if

$$\mathcal{E}(u, v) = \int_K f v \, d\mu, \quad \forall v \in \text{dom}_0 \mathcal{E}$$

As mentioned before we are going to want Δ_μ to be defined as the limit of a discrete operator. We are therefore going to again define a discrete graph Laplacian, the difference is that the following will be without constants.

Definition 2.11: We define the laplacian $\tilde{\Delta}_m$ as

$$\tilde{\Delta}_m = \sum_{y \in V_{m,x}} (f(y) - f(x))$$

Before moving on we will define a harmonic function which will be used extensively in the following proofs.

Definition 2.12: Given m and a point $x \in V_m \setminus V_0$ then define $\psi_x^{(m)}(y) = \delta_{xy}$ for $y \in V_m$.

I might seem that we have only defined $\psi_x^{(m)}$ on V_m , but it is actually defined on the whole of K . It simply follows that we create an harmonic extension of $\psi_x^{(m)}$. We even have that function $\psi_x^{(m)}$ can be used to describe $\tilde{\Delta}_m$.

Proposition 2.13: Let $u \in \text{dom } \mathcal{E}$ then

$$r^{-m} \tilde{\Delta}_m u(x) = \int_K f \psi_x^{(m)} \, d\mu$$

Proof. Since

$$\mathcal{E}_m(u, \psi_x^{(m)}) = r^{-m} \sum_{\substack{p \sim q \\ m}} (u(q) - u(p))(\psi_x^{(m)}(p) - \psi_x^{(m)}(q)) = r^{-m} \sum_{q \in V_{m,x}} (u(x) - u(q)) = r^{-m} \tilde{\Delta}_m u(x)$$

where the second equality comes the fact that only edges containing x are going to see change in $\psi_x^{(m)}$. Now by (2.1) we have that

$$r^{-m} \tilde{\Delta}_m u(x) = \mathcal{E}_m(u, \psi_x^{(m)}) = \mathcal{E}(u, \psi_x^{(m)}) = \int_K f \psi_x^{(m)} \, d\mu$$

as desired. □

Before we begin the proof Δ_μ is the limit of $\tilde{\Delta}_m$ we need some preliminary lemmas.

Lemma 2.14: Let $v : K \rightarrow \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y_i) = 0$ for all $y_i \in V_{m,x}$ then $v(\omega) = 0$ for all $\omega \in \overline{B_x}$ where $B_x = \left(\bigcup_{k>0} S^k(x) \right) \cup \left(\bigcup_i \{y_i\} \right)$

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in B_x$, since the value of the points $S^k(x)$ with $k > 0$ depends on the value of x and its neighbours.

So we need to check the points in $\overline{B_x} \setminus B_x$. Now since B_x is dense in $\overline{B_x}$, then for any $\omega \in \overline{B_x}$ we have that there exists a sequence $x_n \in B_x$ such that $x_n \rightarrow \omega$. By the fact that v is continuous on K it is continuous on $\overline{B_x}$, in which case $|v(x_n) - v(\omega)| \rightarrow 0$ as $n \rightarrow \infty$, but $v(x_n) = 0$ for all x_n , so $v(\omega) = 0$ as desired. $\square$

What lemma 2.14 states, is if any two points in V_m are neighbours, say x and y , and $v(x) = v(y) = 0$, then the function value at any point between x and y is also zero.

Lemma 2.15: Set $F_m = \text{supp } \psi_x^{(m)}$, then

$$F_0 \supset F_1 \supset F_2 \supset F_3 \supset \dots$$

Further F_m is contained in the open ball $B(x, \epsilon_m)$ where $\epsilon_m \rightarrow 0$ as $m \rightarrow \infty$.

Proof. If we set $\Omega_m = \left\{ z \in K : \psi_x^{(m)}(z) = \psi_x^{(m)}(y) = 0, \forall y \in V_{m,z} \right\}$ then $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{B_\alpha} \right)$. It is clear that $\Omega_m \subset \Omega_{m+1}$ and that there are new points in Ω_{m+1} that are not just a point between some already existing points $x_1, x_2 \in \Omega_m$. We therefore have from the way that F_m is constructed that $F_m \supset F_{m+1}$.

The containment of F_m in $B(x, \epsilon_m)$ follows from lemma 2.14 and that $F_m \supset F_{m+1}$. $\square$

We are now ready to prove the pointwise formula

Theorem 2.16: Let $u \in \text{dom } \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m \rightarrow \infty} r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x)$$

holds with the limit uniform across $V_* \setminus V_0$. Conversely, suppose u is a continuous function and the right side of the limit converges uniformly to a continuous function on $V_* \setminus V_0$. Then $u \in \text{dom } \Delta_\mu$ and the equation holds.

Proof. Suppose $u \in \text{dom } \Delta_\mu$ and $\Delta_\mu u = f$, then by proposition 2.13 we have that

$$r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x) = \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu}$$

for any $x \in V_m \setminus V_0$. Uniform convergence of the right side is because of continuity of $f(x)$, so let $\epsilon > 0$ be given, then by lemma 2.15 we can choose m large enough so for all $x, y \in F_m$ that $|f(x) - f(y)| < \epsilon$, then

$$\begin{aligned} \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - f(x) \right| &= \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - \frac{\int_K f(x) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\ &= \left| \frac{\int_K (f(y) - f(x)) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\ &\leq \frac{\int_{F_m} |f(y) - f(x)| d\mu}{\int_K \psi_x^{(m)} d\mu} \\ &\leq \frac{\epsilon \int_{F_m} d\mu}{\int_K \psi_x^{(m)} d\mu} \\ &= \epsilon \end{aligned}$$

in which case

$$\lim_{m \rightarrow \infty} r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x) = \lim_{m \rightarrow \infty} \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} = f(x) = \Delta_\mu u(x)$$

Moving on to the second half of the proof. Suppose that $r^m \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x)$ converges uniformly to a continuous function f . Now for any $v \in \text{dom}_0 \mathcal{E}$ we have

$$\mathcal{E}_m(u, v) = r^{-m} \sum_{\substack{x \sim y \\ m}} (u(y) - u(x)) (v(y) - v(x)).$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x \in V_m \setminus V_0$ then each $v(x)$ will be of the form $-v(x) (u(y) - u(x))$ for each $y \sim_m x$. In which case we get

$$\begin{aligned} \mathcal{E}_m(u, v) &= -r^{-m} \sum_{x \in V_m \setminus V_0} v(x) \left(\sum_{\substack{x \sim y \\ m}} (u(y) - u(x)) \right) \\ &= - \sum_{x \in V_m \setminus V_0} v(x) r^{-m} \tilde{\Delta}_m u(x) \end{aligned}$$

While it seems we lose many terms by removing $v(y)$, but they are instead contained in

$$\sum_{\substack{x \sim y \\ m}} (u(y) - u(x)).$$

Note that the sum is "reset" for every x so we count both the relation $x \sim y$ and $y \sim x$.

Now if we define $f_m(x) = r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x)$ then $f_m \rightarrow f$ uniformly by assumption and we can rewrite the above as

$$\begin{aligned} \mathcal{E}_m(u, v) &= - \sum_{x \in V_m \setminus V_0} v(x) \left(\int_K \psi_x^{(m)} d\mu \right) r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x) \\ &= - \int_K \left(\sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)} \right) d\mu. \end{aligned}$$

Following a quite lengthy argument, we have that

$$\mathcal{E}(u, v) = \lim_{m \rightarrow \infty} \mathcal{E}_m(u, v) = \lim_{m \rightarrow \infty} - \int_K \left(\sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)} \right) d\mu. = - \int_K v(y) f(y) d\mu. \quad (2.3)$$

and we have that $u \in \text{dom} \Delta_\mu$.

We therefore consider

$$\begin{aligned} &\left| \sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) - v(y) f(y) \right| \\ &\leq \left| \sum_{x \in V_m \setminus V_0} (v(x) f_m(x) - v(y) f(y)) \psi_x^{(m)}(y) \right| + \left| \sum_{x \in V_m \setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y) \right| \end{aligned}$$

Looking at each term individually we first see that

$$\begin{aligned} \left| \sum_{x \in V_m \setminus V_0} (v(x)f_m(x) - v(y)f(y)) \psi_x^{(m)} \right| \\ \leq \sum_{x \in V_m \setminus V_0} (|v(x)f_m(x) - v(y)f_m(y)| + |v(y)f_m(y) - v(y)f(y)|) \psi_x^{(m)}(y) \end{aligned}$$

now again we will look at each term starting with $|v(y)f_m(y) - v(y)f(y)|$. Here

$$|v(y)f_m(y) - v(y)f(y)| = |v(y)| |f_m(y) - f(y)|$$

and since v is continuous on K , which is compact, there is a constant C_1 so $|v(y)| \leq C_1$ for all $x \in K$. Since f_m converges uniformly to f we can choose m_1 such that $|f_m(y) - f(y)| \leq \frac{\epsilon}{C_1}$ and the convergence of that term follows.

Next is $|v(x)f_m(x) - v(y)f_m(y)|$, here we find

$$|v(x)f_m(x) - v(y)f_m(y)| \leq |v(x)| |f_m(x) - f(y)| + |f_m(y)| |v(x) - v(y)|$$

First $|f_m(y)| |v(x) - v(y)|$, here $|f_m(y)| \leq C_2$ is uniformly bounded by some C_2 . By lemma 2.15 and continuity of v we can choose m_2 so $|v(x) - v(y)| \leq \frac{\epsilon}{C_2}$ for all $x \in F_{m_2}$. The reason we can choose m_2 is because $\psi_x^{(m)}$ is multiplied onto all the terms so we just restrict v to F_{m_2} .

Second $|v(x)| |f_m(x) - f(y)|$, again $|v(x)| \leq C_1$, but

$$|f_m(x) - f(y)| \leq |f_m(x) - f(x)| + |f(x) - f_m(y)| \leq |f_m(x) - f(x)| + |f(x) - f(y)| + |f(y) - f_m(y)|$$

First choose m_3 so $|f_{m_3}(x) - f(x)| \leq \frac{\epsilon}{3C_1}$, next by continuity of f choose m_4 so $|f(x) - f(y)| \leq \frac{\epsilon}{3C_1}$ for all $x \in F_{m_4}$, lastly choose m_5 so $|f(y) - f_{m_5}(y)| \leq \frac{\epsilon}{3C_1}$.

So we now have for $m \geq \max\{m_1, m_2, \dots, m_5\}$ that

$$\left| \sum_{x \in V_m \setminus V_0} (v(x)f_m(x) - v(y)f(y)) \psi_x^{(m)} \right| \leq \epsilon \sum_{x \in V_m \setminus V_0} \psi_x^{(m)} \leq \epsilon \sum_{x \in V_m \setminus V_0} 1 \leq \epsilon.$$

We are now going to look at $\left| \sum_{x \in V_m \setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y) \right|$ here

$$\left| \sum_{x \in V_m \setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y) \right| = |v(y)f(y)| \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right|$$

Now to show $\left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right| = 0$ as $m \rightarrow \infty$, we start by looking at $y \in V^*$. Choose m_6 so $y \in V_{m_6}$.

Next assume $y \in K$, then since V^* is dense in K we can choose a sequence $(x_n)_{n=1}$ where $x_n \rightarrow y$ as $n \rightarrow \infty$, and since $\psi_x^{(m)}$ is continuous on K then $\psi_x^{(m)}(x_n) \rightarrow \psi_x^{(m)}(y)$ as $n \rightarrow \infty$. This gives us the following

$$\left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right| \leq \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) \right| + \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) - 1 \right|$$

Now choose an n_1 so $|\psi_x^{(m)}(x_{n_1}) - \psi_x^{(m)}(y)| \leq \epsilon$ and an m_7 so $x_{n_1} \in V_{m_7}$. Both can then be bounded by $m \geq \max\{n_1, m_7\}$. Thus we have the convergence of (2.3). $\square$

2.2 And back again

We will now extend the Kigami Laplacian to the Dirichlet Laplacian, and we will be following [BC22] and [FS92] very closely in defining the Dirichlet Laplacian

This will be done through Dirichlet forms, thus we begin with them.

Definition 2.17: (i) A *quadratic form* $\mathcal{E}$ on $\mathcal{H}$ is a non-negative, definite, symmetric bilinear form from $\text{dom } \mathcal{E} \times \text{dom } \mathcal{E} \rightarrow \mathbb{R}$, where $\text{dom } \mathcal{E}$ is a dense subspace $\mathcal{H}$.

(ii) A quadratic form is said to be closed if $\text{dom } \mathcal{E}$ equipped with the norm

$$\|f\|_{\text{dom } \mathcal{E}}^2 = \|f\|^2 + \mathcal{E}(f, f)$$

is a Hilbert space.

(iii) A quadratic form $\mathcal{E}$ on $\mathcal{H}$ is said to be closable if it admits a closed extension. That is there exists a closed quadratic form $\bar{\mathcal{E}}$ such $\text{dom } \mathcal{E} \subset \text{dom } \bar{\mathcal{E}}$ and $\bar{\mathcal{E}}$ coincides with $\mathcal{E}$ on $\text{dom } \mathcal{E} \times \text{dom } \mathcal{E}$

(iv) We further call a quadratic form $\mathcal{E}$ a Dirichlet form if it has the following property. If given $u, v \in \text{dom } \mathcal{E}$ so that for almost every $x, y \in K$ we have

$$|v(x) - v(y)| \leq |u(x) - u(y)|, \quad \text{and} \quad |v(x)| \leq |u(x)|$$

then

$$\mathcal{E}(v, v) \leq \mathcal{E}(u, u).$$

Now recall from definition 2.8 that

$$\mathcal{E}(f, f) = \lim_{m \rightarrow \infty} \mathcal{E}_m(f, f)$$

with domain $\text{dom}_0 \mathcal{E}$ where

$$\mathcal{E}_m(f, f) = \sum_{\substack{x \sim y \\ m}} (f(x) - f(y))^2$$

By [FS92, theorem 4.1] we have that $(\mathcal{E}, \text{dom}_0 \mathcal{E})$ is a Dirichlet form on $L^2(K, \mu)$. Then by [Bau25, Lemma 1.3.3] we have that

$$\mathcal{E}(f, f) = - \int_K \Delta f g \, d\mu$$

where Δ is the unique unique densely defined self-adjoint operator which is an extension of the Kigami Laplacian Δ_μ . The operator Δ with domain $\text{dom } \Delta$ is referred to as the Dirichlet Laplacian. **Something feels wrong, i should use 1.3.5 but i end up using 1.3.3. But then i need that the kigami laplacian is densely defined and self adjoint and so forth.**

2.3 Discrete Fractional Gaussian Fields

Recalling remark 2.2 then we consider the eigenvalues $0 < \lambda_1^m \leq \lambda_2^m \leq \dots \leq \lambda_{N_m}^m$ of $-\Delta_m$. Next let $(\Phi_i^m)_{1 \leq i \leq N_m}$ be the corresponding eigenfunctions with respect to the measure μ_m . This leads us to defining the following. **Still something about eigenfunctions**

Definition 2.18: Let $s \geq 0$ we define the *discrete Riesz kernel* on V_m as

$$G_s^m(x, y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \phi_i^m(y), \quad \text{for } x, y \in V_m$$

We can further by construction see that the matrix $(G_s^m(x, y))_{x, y \in V_m}$ is symmetric and is non-negative. **I cannot figure it out, write stuff here later**, thus it can be the covariance matrix of a Gaussian vector.

Having constructed a discrete Laplacian we continue with constructing another Laplacian.

Definition 2.19: For $f \in \ell(V_m)$ and $s \geq 0$, the discrete fractional Laplacian $(-\Delta_m)^s : L^2(V_m, \mu_m) \rightarrow L^2(V_m, \mu_m)$ is defined by

$$(-\Delta_m)^{-s} f(x) = \frac{1}{a_m} \sum_{y \in V_m} G_s^m(x, y) f(y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y).$$

Remark: The attentive reader will have noticed that $\frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) = \langle \phi_i^m, f \rangle_{L^2(V_m, \mu_m)}$ and therefore that $(-\Delta_m)^{-s} f(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \langle \phi_i^m, f \rangle_{L^2(V_m, \mu_m)}$

Having stated the discrete fractional Laplacian we move on to proving some useful properties of the discrete fractional Laplacian.

Proposition 2.20:

- (i) $(-\Delta_m)^{-s}$ is a uniformly bounded operator
- (ii) $(-\Delta_m)^{-s}$ is a symmetric operator.

Proof. We start with (i). We have

$$\begin{aligned} \|(-\Delta_m)^{-s} f(x)\|_{L^2(V_m, \mu_m)} &\leq \left\| \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) \right\|_{L^2(V_m, \mu_m)} \\ &\leq \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \frac{1}{a_m} \left\| \phi_i^m(x) \sum_{y \in V_m} \phi_i^m(y) f(y) \right\|_{L^2(V_m, \mu_m)} \\ &\leq (\lambda_1^m)^{-s} \|f\|_{L^2(V_m, \mu_m)} \end{aligned}$$

where the last inequality comes from the fact that $0 < \lambda_1 < \lambda_2 < \dots$. The uniform bound comes from the fact that $\inf_m \lambda_1^m > 0$. This follows from [FS92, lemma 5.2], since if $\inf_m \lambda_1^m \leq 0$ the series in the lemma would not converge.

Moving on to (ii), we have that

$$\begin{aligned} \langle f, (-\Delta_m)^{-s} g \rangle &= \int_{V_m} f(-\Delta_m)^{-s} g \, d\mu_m(x) \\ &= \int_{V_m} f(x) \cdot \left(\sum_{i,j=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) g(y) \right) d\mu_m(x) \\ &= \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \frac{1}{a_m} \sum_{y \in V_m} \phi_j^m(y) g(y) \int_{V_m} \left(\sum_{i=1}^{N_m} \phi_i^m(x) \langle \phi_i^m, f \rangle \right) \phi_j^m(x) d\mu_m(x) \\ &= \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \frac{1}{a_m} \sum_{y \in V_m} \phi_j^m(y) g(y) \int_{V_m} \phi_j^m(x) \langle \phi_j^m, f \rangle \phi_j^m(x) d\mu_m(x) \\ &= \sum_{j=1}^{N_m} \left[\int_{V_m} \left((\lambda_j^m)^{-s} \phi_j(x) \langle \phi_j^m, f \rangle \right) \phi_j^m(x) \langle \phi_j^m, g \rangle d\mu_m(x) \right. \\ &\quad \left. + \int_{V_m} \left((\lambda_j^m)^{-s} \phi_j^m(x) \langle \phi_j^m, f \rangle \right) \left(\sum_{i=1, i \neq j}^{N_m} \phi_i^m(x) \langle \phi_i^m, g \rangle \right) d\mu_m(x) \right] \\ &= \int_{V_m} (-\Delta_m)^{-s} f g \, d\mu_m(x) \\ &= \langle (-\Delta_m)^{-s} f, g \rangle \end{aligned}$$

using that $f = \sum_{i=1}^{N_m} \phi_i \langle \phi_i, f \rangle$ and $\int_{V_m} \phi_i \phi_j \, d\mu_m$ is 1 if $j = i$ and 0 if $i \neq j$. $\square$

Since we have now described the discrete fractional laplacian we will move on to the continuous fractional Laplacian.

Semi-groups

When stating the Fractional Laplacian one use that it admits a heat kernel through a semigroup generated by $-\Delta$. We will therefore first define a semigroup along with showing that $-\Delta$ generates a semigroup and define what the heat kernel is.

Definition 2.21: Let $\mathcal{H}$ be a Hilbert space. A strongly continuous self-adjoint contraction semigroup is a family of self-adjoint operators $(P_t)_{t \geq 0} : \mathcal{H} \rightarrow \mathcal{H}$

- (i) For $s, t \geq 0$, $P_t \circ P_s = P_{s+t}$ (semi-group property).
- (ii) For every $f \in \mathcal{H}$, $\lim_{t \rightarrow 0} P_t f = f$ (strong continuity).
- (iii) For every $f \in \mathcal{H}$ and $t \geq 0$, $\|P_t f\| \leq \|f\|$ (contraction property).

The next theorem also holds if we are given $(P_t)_{t \geq 0}$ as a strongly continuous self-adjoint contraction semigroup, then there exists a self-adjoint, non-positive and densely defined operator that generates $(P_t)_{t \geq 0}$, but we only need the the operator to generate the semi-group.

Theorem 2.22: If A is a densely defined self adjoint operator on $\mathcal{H}$ then it is the generator of the strongly continuous self-adjoint contraction semigroup on $\mathcal{H}$ defined as $P_t = e^{-tA}$.

Proof. Let A be given as stated. Then from the spectral theorem [Bau25, theorem 1.1.2] and there is a measure space (ω, ν) and a unitary map $U : L^2(\Omega, \nu) \rightarrow \mathcal{H}$ and a non negative real valued measurable function λ on ω such that

$$U^{-1} A U f(x) = -\lambda(x) f(x)$$

for $x \in \Omega, Uf \in \text{dom } A$. Then from the functional calculus [Bau25, theorem 1.1.3] we define $P_t : \mathcal{H} \rightarrow \mathcal{H}$ such that

$$U^{-1} P_t U f(x) = e^{-t\lambda(x)} f(x)$$

Is enough to show Q_t is all these things since $P_t = U e^{-\lambda t} U^{-1}$ is still all the things? by def. that unitary maps are surjective. We will prove that $Q_t = U^{-1} P_t U$ with domain $\text{dom } Q_t$ is a self-adjoint strongly continuous contraction semigroup on $L^2(\Omega, \nu)$ with generator A .

Starting with generator, we see that

$$\begin{aligned} \lim_{t \rightarrow 0} \left\| \frac{Q_t f - f}{t} - A f \right\| &= \lim_{t \rightarrow 0} \sqrt{\left\langle \frac{Q_t f - f}{t} - A f, \frac{Q_t f - f}{t} - A f \right\rangle} \\ &= \lim_{t \rightarrow 0} \sqrt{\left\langle \frac{Q_t f - f}{t}, \frac{Q_t f - f}{t} - A f \right\rangle - \left\langle A f, \frac{Q_t f - f}{t} - A f \right\rangle} \\ &\leq \lim_{t \rightarrow 0} \sqrt{\left\langle \frac{Q_t f - f}{t}, \frac{Q_t f - f}{t} - A f \right\rangle} \\ &= \lim_{t \rightarrow 0} \sqrt{\int_{\Omega} \left(\frac{Q_t f - f}{t} \right) \left(\frac{Q_t f - f}{t} - A f \right) d\nu} \\ &= \lim_{t \rightarrow 0} \sqrt{\int_{\Omega} \left(\frac{(e^{-\lambda t} - 1)f}{t} \right) \left(\frac{Q_t f - f}{t} - A f \right) d\nu} \\ &= 0 \end{aligned}$$

where the convergence comes from the fact that $e^{-\lambda t}$ converges to 1 much faster than $1/t$ diverges to ∞ . Thus A is a generator.

Next we have that Q_t is self-adjoint, for which we start with symmetry

$$\langle Q_t f, g \rangle = \int_{\Omega} Q_t f g d\nu = \int_{\Omega} e^{-\lambda t} f g d\nu = \langle f, Q_t g \rangle.$$

Next we have that $\text{dom } Q_t$ is dense in $L^2(\Omega, \nu)$ since if $f_n \rightarrow f$ in $L^2(\omega, \nu)$ and $Q_t f_n \rightarrow g$ then

$$\lim_{n \rightarrow \infty} Q_t f_n = \lim_{n \rightarrow \infty} e^{-\lambda t} f_n = e^{-\lambda t} f = Q_t f.$$

Now showing that $\text{dom}(Q_t)^* = \text{dom } Q_t$, then since Q_t is symmetric we have that

$$|\langle f, Q_t g \rangle| = |\langle Q_t f, g \rangle| \leq \|Q_t f\| \|g\|.$$

I think this arguments holds. see just below definition of self-adjoint. Thus we have Q_t is self-adjoint. To show the semigroup properties, we have (i)

$$Q_{t+s}f = e^{-\lambda(t+s)}f = e^{-\lambda t}e^{-\lambda s}f = Q_t Q_s f$$

further for (ii) we have

$$\lim_{t \rightarrow \infty} Q_t f = \lim_{t \rightarrow \infty} e^{-t\lambda} f = f$$

and lastly (iii)

$$\|Q_t f\| = \int_{\Omega} (e^{-\lambda t} f)^2 d\nu \leq \int_{\Omega} f^2 d\nu = \|f\|.$$

Thus we have that Q_t is a self-adjoint strongly continuous contraction semigroup on $L^2(\Omega, \nu)$ with generator A . $\square$

With semigroups out of the way we can define the heat kernel.

Definition 2.23: Let $(P_t)_{t \geq 0}$ be a strongly continuous self-adjoint contraction semigroup on $L^2(X, \mu)$. We say that the semigroup $(P_t)_{t \geq 0}$ admits a heat kernel if the heat kernel measure have a density with respect to μ , i.e. there exists a measurable function $p : \mathbb{R}_{>0} \times X \times X \rightarrow \mathbb{R}_{\geq 0}$ such that for every $t > 0$, almost everywhere we have

$$P_t f(x) = \int_K p_t(x, y) f(y) d\mu(y)$$

with $x, y \in X, f \in L^\infty(X, \mu)$.

Fractional Laplacian

Before we can define the continuous fractional Laplacian, we need to define the eigenvalues and eigenfunctions for $-\Delta$.

Recall that the Laplacian Δ with domain $\mathcal{D}(\Delta)$ is a densely defined self-adjoint operator. Therefore by theorem 2.22 it is the generator of a strongly continuous Markov semigroup $\{P_t\}_{t \geq 0}$ on $L^2(K, \mu)$.

This semigroup admits by definition 2.23 a heat kernel $p_t(x, y), t > 0, x, y \in K$, wrt. the normalized Hausdorff measure μ . The heat kernel $p_t(x, y)$ is called the Dirichlet heat kernel on K . Further this heat kernel satisfies that for some $c_1, c_2 \in (0, \infty)$ find ref

$$p_t(x, y) \leq c_1 t^{-\frac{d_h}{d_w}} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) \quad (2.4)$$

for every $x, y \in K \times K$ and $t \in (0, \infty)$ where d_h is the Hausdorff dimension and d_w is the walk dimension.

The Dirichlet heat kernel is quite central in the definition of the continuous fractional Laplacian, as its bounds and definition will be used quite a bit throughout the proofs of this next section. One of the more important properties of the heat kernel is that it can be described as a sum of eigenvalues and eigenfunctions of Δ . The following theorem gives us the existence of the eigenvalues, functions and the sum representation of the Dirichlet heat kernel.

Theorem 2.24: There exists a an orthonormal basis $\{\phi_i\}_{i=1}^\infty$ of $L^2(K, \mu)$ with ϕ_j being eigenfunctions of $-\Delta$. If λ_i is the eigenvalue belonging to ϕ_i , then $\lambda_i > 0$ and $\lim_{i \rightarrow \infty} \lambda_i = \infty$. Moreover

$$p_t(x, y) = \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y)$$

where the convergence is uniform for all $t \in (0, \infty)$ and $x, y \in K \times K$

A detailed proof can be found in [Dod81, Theorem 2.2].

The reason that we find eigenfunctions for $-\Delta$ is such that the eigenfunctions satisfy

$$\Delta\phi_j = -\lambda_j\phi_j$$

The eigenvalues cannot only be used to represent the heat kernel but will also be used to represent the fractional Laplacian, and the following lemma gives us an idea of what parameter values gives convergence.

Lemma 2.25: Let $0 < \lambda_1 \leq \lambda_2 < \dots$ be the eigenvalues of $-\Delta$

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty. \iff s > \frac{d_h}{2d_w}$$

Proof. This follows from [Kig01, lemma 5.1.3] which states that there exists a constant $c_1, c_2 > 0$ such that

$$c_1 \cdot n^{2/d_s} \leq \lambda_n \leq c_2 \cdot n^{2/d_s} \quad (2.5)$$

Assume $s > \frac{d_h}{2d_w}$ then by (2.5)

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} \leq \sum_{j=1}^{\infty} \frac{1}{(c_1 n^{2/d_s})^{2s}} < \infty$$

since

$$\frac{4s}{d_s} > \frac{\frac{2d_h}{d_w}}{\frac{2d_h}{d_w}} = 1$$

the series is convergent.

Assume now that

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty.$$

then again by (2.5) we have that

$$\sum_{j=1}^{\infty} \frac{1}{(c_2 n^{2/d_s})^{2s}} \leq \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty$$

and the left sum only converges if $4s/d_s > 1$ in which case $s > d_s/4 = d_h/2d_w$ as desired. $\square$

Thus we are ready to define the fractional Laplacian.

Definition 2.26: Let $s \geq 0$, for $f \in L^2(K, \mu)$ the fraction Laplacian $(-\Delta)^{-s}$ is defined as

$$(-\Delta)^{-s}f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y) \, d\mu(y)$$

Remark: Ones again we notice that $\langle \phi_i, f \rangle = \int_K \phi_i f \, d\mu$, and can rewrite $(-\Delta)^{-s}f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle$.

Now that we have the smooth fractional Laplacian defined we will show some of its very useful properties.

Proposition 2.27: Let $(-\Delta)^{-s}$ be defined as in definition 2.26 then:

- (i) $(-\Delta)^{-s}$ is a linear operator.
- (ii) If ϕ_n be an orthonormal eigenfunction, then $(-\Delta)^{-s}\phi_n = \frac{1}{\lambda_n^s}\phi_n$.
- (iii) $(-\Delta)^{-s}$ is a bounded operator.
- (iv) $(-\Delta)^{-s}$ is a symmetric operator.

Proof. We start with the proof of (i) Let $f, g \in L^2(K, \mu)$ and α some constant, then by definition of $(-\Delta)^{-s}$ we have

$$\begin{aligned} (-\Delta)^{-s}(\alpha f + g) &= \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) (\alpha f(y) + g(y)) \, d\mu(y) \\ &= \alpha \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y) \, d\mu(y) + \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) g(y) \, d\mu(y) \\ &= \alpha (-\Delta)^{-s} f + (-\Delta)^{-s} g \end{aligned}$$

Moving on to (ii). We note that since

$$\int_K \phi_j \phi_i \, d\mu = \delta_{ij}$$

where δ_{ij} is the Kronecker delta. So for any eigenfunction ϕ_n we have

$$(-\Delta)^{-s} \phi_n = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j(x) \int_K \phi_j(y) \phi_n(y) \, d\mu(y) = \frac{1}{\lambda_n^s} \phi_n.$$

Next we have (iii). This follows from

$$\begin{aligned} \|(-\Delta)^{-s} f\|^2 &= \left\| \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right\|^2 \\ &= \left\langle \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle, \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right\rangle \\ &= \sum_j \left| \frac{1}{\lambda_j^s} \langle \phi_j, f \rangle \right|^2 \langle \phi_j, \phi_j \rangle \\ &\leq \left(\frac{1}{\lambda_1^s} \right)^2 \sum_{j=1}^{\infty} |\langle \phi_j, f \rangle|^2 \\ &= \left(\frac{1}{\lambda_1^s} \right)^2 \|f\|^2 \end{aligned}$$

in which case we have our bound. Lastly we will show (iv)

$$\begin{aligned}
 \langle (-\Delta)^{-s} f, g \rangle &= \int_K \left(\sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right) g \, d\mu \\
 &= \int_K \left(\sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right) \left(\sum_{i=1}^{\infty} \phi_i \langle \phi_i, g \rangle \right) d\mu \\
 &= \sum_{j=1}^{\infty} \int_K \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \left(\sum_{i=1}^{\infty} \phi_i \langle \phi_i, g \rangle \right) d\mu \\
 &= \sum_{j=1}^{\infty} \int_K \frac{1}{\lambda_j^s} \phi_j^2 \langle \phi_j, f \rangle \langle \phi_j, g \rangle d\mu \\
 &= \sum_{j=1}^{\infty} \left[\int_K \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \phi_j \langle \phi_j, g \rangle d\mu + \int_K \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, g \rangle \left(\sum_{i=1, i \neq j}^{\infty} \phi_i \langle \phi_i, f \rangle \right) d\mu \right] \\
 &= \sum_{j=1}^{\infty} \int_K \left(\frac{1}{\lambda_j^s} \phi_j \langle \phi_j, g \rangle \right) \left(\sum_{i=1}^{\infty} \phi_i \langle \phi_i, f \rangle \right) d\mu \\
 &= \int_K \left(\sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, g \rangle \right) f \, d\mu \\
 &= \langle f, (-\Delta)^s g \rangle
 \end{aligned}$$

as desired. $\square$

While the definition of the fractional Laplacian is not the most complicated, we are still able to show an equality between $(-\Delta)^{-s}$ and the Riesz kernel. We are therefore able to use the estimates of the heat kernel $p_t(x, y)$ in conjunction with $(-\Delta)^{-s}$.

Definition 2.28: For parameter $s \geq 0$ we define the fractional Riesz kernel G_s by

$$G_s(x, y) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} p_t(x, y) \, dt, \quad x, y \in K, x \neq y$$

where Γ denotes the gamma function.

Remark: Note that it follows from (2.4) that $G_s(x, y)$ actually converges for every $s \geq 0$ where $x \neq y$.

We will now show some of the bounds of the fractional Riesz kernel.

Proposition 2.29: 1. If $s \in [0, d_h/d_w)$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x, y) \leq \frac{C}{d(x, y)^{d_h - s d_w}}$$

2. If $s = d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x, y) \leq C |\ln d(x, y)|$$

3. If $s > d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x, y) \leq C$$

Before stating this proof we will quickly state a lemma needed

Lemma 2.30: There exists a constant $C > 0$ such that for every $x, y \in K$ and $t \geq 1$,

$$|p_t(x, y)| \leq C e^{-\lambda_1 t}$$

where $\lambda_1 > 0$ is the first non-zero eigenvalue of $-\Delta$.

Proof. We have from theorem 2.24 that $p_t(x, y)$ has the expansion

$$p_t(x, y) = \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y). \quad (2.6)$$

Since ϕ_j are eigenvalues, we have from definition of p_t , that for any $t > 0$ that

$$\phi_j(x) = e^{\lambda_j t} \int_K p_t(x, y) \phi_j(y) \, d\mu(y).$$

then from Cauchy-Schwarz inequality, we have for every $t > 0$

$$|\phi_j(x)| \leq e^{\lambda_j t} \left(\int_K p_t(x, y)^2 \, d\mu(y) \right)^{\frac{1}{2}} \left(\int_K \phi_j(y)^2 \, d\mu(y) \right)^{\frac{1}{2}} = e^{\lambda_j t} p_{2t}(x, x)^{1/2}$$

where the last equality comes from the fact that $\int_K \phi_i(y) \phi_j(y) \, d\mu = \delta_{ij}$ so $\int_K \phi_j(y)^2 \, d\mu(y) = 1$. That combined with (2.6) gives us $\int_K p_t(x, y)^2 \, d\mu(y) = \int_K p_{2t}(x, y) \, d\mu(y)$ which gives us the equality.

Now choosing $t = 1/4$ we get along with (2.4) that there is a constant $C > 0$ so for every $x \in K$ we have

$$|\phi_j| \leq C e^{\lambda_j/4}.$$

Returning to (2.6) we get for every $x, y \in K$ and $t \geq 1$ that

$$\begin{aligned} |p_t(x, y)| &\leq \sum_{j=1}^{\infty} e^{-\lambda_j t} |\phi_j(x)| |\phi_j(y)| \\ &\leq C^2 \sum_{j=1}^{\infty} e^{-\lambda_j t} e^{\lambda_j/2} \\ &= C^2 \sum_{j=1}^{\infty} e^{-\lambda_j t} e^{\lambda_j/2} e^{\lambda_j} e^{-\lambda_j} \\ &\leq C^2 e^{-\lambda_1 t} e^{\lambda_1} \sum_{j=1}^{\infty} e^{-\lambda_j/2} \end{aligned}$$

here the last inequality comes from the fact that $0 < \lambda_1 < \lambda_2 < \dots$ and $t \geq 1$ so $e^{\lambda_j(1-t)} \leq e^{\lambda_1(1-t)}$. Thus we have the desired bound. $\square$

We are now ready to prove:

Proof of proposition 2.29. We have that

$$\begin{aligned} G_s(x, y) &= \frac{1}{\Gamma(s)} \int_0^{\infty} t^{s-1} p_t(x, y) \, dt \\ &= \frac{1}{\Gamma(s)} \int_0^1 t^{s-1} p_t(x, y) \, dt + \frac{1}{\Gamma(s)} \int_1^{\infty} t^{s-1} p_t(x, y) \, dt \end{aligned}$$

The integral $\frac{1}{\Gamma(s)} \int_1^{\infty} t^{s-1} p_t(x, y) \, dt$ can be uniformly bounded by a constant using lemma 2.30. Thus we only need to bound the integral $\int_0^1 t^{s-1} p_t(x, y) \, dt$. By (2.4) we have that

$$\int_0^1 t^{s-1} p_t(x, y) \, dt \leq c_1 \int_0^1 t^{s-1-\frac{d_h}{d_w}} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) \, dt.$$

Since the bound of the integral relies on the value of s we will split it up into the cases of the proposition.

Let $s > d_h/d_w$ then we simply have

$$\int_0^1 t^{s-1-\frac{d_h}{d_w}} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) \, dt \leq \int_0^1 t^{s-1-\frac{d_h}{d_w}} \, dt.$$

Next if $s < d_h/d_w$, then using a change of variable $t = ud(x, y)^{d_w}$, we find that

$$\begin{aligned} & \int_0^1 t^{s-1-\frac{d_h}{d_w}} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) dt \\ &= d(x, y)^{sd_w-d_h} \int_0^{\frac{1}{d(x, y)^{d_w}}} u^{s-1-\frac{d_h}{d_w}} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w+1}} \right) du \\ &\leq d(x, y)^{sd_w-d_h} \int_0^\infty u^{s-1-\frac{d_h}{d_w}} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w+1}} \right) du. \end{aligned}$$

convergence of the last integral comes from the fact that $e^{-c_2(1/u)^{1/d_w}}$ converges faster than $1/u$ diverges. Lastly for $s = d_h/d_w$, we again use the change of variable $t = ud(x, y)^{d_w}$ and setting $R = \text{diam}(K)$ and note that $0 < 1/R^{d_w} < 1/d(x, y)^{d_w}$, thus we find

$$\begin{aligned} & \int_0^1 t^{s-1-\frac{d_h}{d_w}} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) dt \\ &= \int_0^{\frac{1}{d(x, y)^{d_w}}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w+1}} \right) du \\ &\leq \int_{\frac{1}{R^{d_w}}}^{\frac{1}{d(x, y)^{d_w}}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w+1}} \right) du + \int_0^{\frac{1}{R^{d_w}}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w+1}} \right) du \\ &\leq \int_{\frac{1}{R^{d_w}}}^{\frac{1}{d(x, y)^{d_w}}} u^{-1} du + \int_0^{\frac{1}{R^{d_w}}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w+1}} \right) du \\ &\leq d_w |\ln d(x, y)| + \int_0^{\frac{1}{R^{d_w}}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w+1}} \right) du \\ &\leq C |\ln d(x, y)|. \end{aligned}$$

Thus we have our desired bounds. $\square$

2.4 Bounds of the fractional Laplacian

In the following we will prove some bounds for the fractional Laplacian. Since the goal of this thesis is to prove convergence of fractional Gaussian fields within a specific space of test functions, we will only prove the lemmas for that space. The space of functions is given by

Definition 2.31: We consider the following space of test functions

$$S(K) = \left\{ f \in C_0(K) : \forall k \geq 0 \lim_{n \rightarrow \infty} n^k \left| \int_K \phi_n(y) f(y) d\mu(y) \right| = 0 \right\}$$

We consider this space with the topology defined by the family of norms

$$\|f\|_k = \left\| (-\Delta)^{-k} f \right\|_{L^2(K, \mu)}.$$

By [Pie66, Theorem 2] and lemma 2.25 we have that $S(K)$ is a Fréchet Nuclear space. That a space is *Fréchet* means that it is nuclear and the topology is given på a countable family of semi-norms. We will quickly define a nuclear space, but will not delve into the subject matter. The following definition is from [Trè67, Chapter 50].

Definition 2.32: Let E be a locally convex Hausdorff space and $\|\cdot\|_p$ a continuous semi-norm on E . Further let $\hat{E}_p$ a Banach space defined to be the completion of the normed space $E \setminus \ker \|\cdot\|_p$. Then E is a *nuclear space* if to every continuous semi-norm $\|\cdot\|_p$ on E there exists another continuous semi-norm on E with $\|\cdot\|_q \geq \|\cdot\|_p$ such that the canonical mapping $\hat{E}_p \rightarrow \hat{E}_q$ is *nuclear*.

A mapping $u : E \rightarrow F$, where E, F are Banach spaces, is *nuclear* if there is a sequence in the closed unit ball of E' , $\{x'_k\}$, a sequence in the closed unit ball of F , $\{y_k\}$ and a complex sequence $\{\lambda_k\}$ with $\sum_k |\lambda_k| < \infty$ such that $u(x) = \sum_{i=1}^{\infty} \lambda_i x'_i(x) y_i$ for all $x \in E$. Furthermore, the trace-norm $\|u\|_{\text{Tr}}$ is equal to the infimum of the numbers $\sum_k |\lambda_k|$ over the set of all representations of u as such a series.

Remark: We have just defined a nuclear map on a Banach spaces, but do note that the definition can be generalized to locally convex Hausdorff spaces.

As we can see the concepts of a nuclear spaces is quite complicated, but its properties will be very useful. It will allow us to prove convergence of Gaussian fields through the characteristics function and existence of Fractional Gaussian fields through a functional. **Is this okay to write?** But first we will prove some lemmas regarding the fractional Laplacian and the fractional Riesz kernel.

Lemma 2.33: Let $s > 0$. For $f \in S(K)$ and $x \in K$

$$(-\Delta)^{-s} f = \int_K G_s(x, y) f(y) \, d\mu(y)$$

Proof. We will first make sure the integral on the left side actually converges. But this holds because $S(K) \subset C(K)$ so f will be bounded on K . Thus we just need that for every $x \in K$ $\int_K G_s(x, y) \, d\mu(y)$ is convergent. for $s \in [d_h/d_w, \infty)$ the convergence follows from proposition 2.29. Lastly for $s \in (0, d_h/d_w)$ then $d_h - sd_w < d_h$. If we recall proposition 1.13 we can

$$\int_K G_s(x, y) \, d\mu(y) \leq \int_K \frac{1}{d(x, y)^{d_h - sd_w}} \, d\mu(y) < \infty$$

which follows from the Alfons regular Hausdorff measure **why????? Choosing r so the ball $B(x, r) \supset d(x, y)^{d_h - sd_w}$ But then how does this relate to $\mu(B(x, r))$.**

Now using Fubini's theorem, theorem 2.24 and the definition of G_s we get

$$\begin{aligned} \int_K G_s(x, y) f(y) \, d\mu(y) &= \int_K \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} p_t(x, y) \, dt f(y) \, d\mu(y) \\ &= \int_K \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \sum_{j=1}^\infty e^{-\lambda_j t} \phi_j(x) \phi_j(y) \, dt f(y) \, d\mu(y) \\ &= \sum_{j=1}^\infty \phi_j(x) \int_K \phi_j(y) f(y) \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, dt \, d\mu(y) \\ &= \sum_{j=1}^\infty \frac{1}{\lambda_j^s} \phi_j(x) \int_K \phi_j(y) f(y) \, d\mu(y) \\ &= (-\Delta)^{-s} f \end{aligned}$$

where we use that $\frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, dt = -\lambda_j^{-s}$. This follows from if we use the substitution $u = \lambda_j t$ we get

$$\frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, dt = \frac{1}{\Gamma(s) \cdot \lambda_j^s} \int_0^\infty u^{s-1} e^{-u} \, du = \frac{1}{\Gamma(s) \lambda_j^s} \Gamma(s) = \lambda_j^{-s}.$$

□

An immediate corollary follows:

Corollary 2.34: Let $s > 0$, for $f, g \in S(K)$

$$\int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu = \int_K \int_K f(x) g(y) G_{2s}(x, y) \, d\mu(x) \, d\mu(y)$$

Proof. Now from the definition of $(-\Delta)^{-s}$, proposition 2.27 and the previous lemma, we find that

$$\begin{aligned}
 \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu &= \langle (-\Delta)^{-s} f, (-\Delta)^{-s} g \rangle \\
 &= \langle f, (-\Delta)^{-2s} g \rangle \\
 &= \int_K f (-\Delta)^{-2s} g \, d\mu \\
 &= \int_K \int_K f(x) g(y) G_{2s}(x, y) \, d\mu(x) \, d\mu(y)
 \end{aligned}$$

□

Chapter 3

Fractional Gaussian Fields

In this chapter we will be constructing the fractional Gaussian fields, that we will later see converge. The construction of the fractional Gaussian fields very much depends on the Riesz kernel and the Dirichlet Laplacian.

We will also give an example of an discrete fractional Gaussian fields of which we have constructed some simulations.

3.1 Discrete fractional Gaussian fields

Definition 3.1: Let $s \geq 0$, then a *discrete fractional Gaussian field* X_s^m with parameter s on V_m is a Gaussian vector indexed by V_m with mean zero and covariance matrix $\Sigma = (G_{2s}^m(x, y))_{x, y \in V_m}$

For any $f \in \ell(V_m)$ we will use the notation

$$X_s^m(f) = \frac{1}{a_m} \sum_{p \in V_m} f(p) X_s^m(p)$$

We can then note that the discrete fractional Gaussian field has the following property

Proposition 3.2: For $f, g \in \ell(V_m)$ we have that

$$\mathbb{E} [X_s^m(f) X_s^m(g)] = \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p)$$

Proof. Using proposition 2.20 we find

$$\begin{aligned} \mathbb{E} [X_s^m(f) X_s^m(g)] &= \frac{1}{2a_m} \sum_{p, q \in V_m} f(p) g(q) \mathbb{E} [X_s^m(p) X_s^m(q)] \\ &= \frac{1}{2a_m} \sum_{p, q \in V_m} f(p) g(q) G_{2s}^m(p, q) \\ &= \frac{1}{a_m} \sum_{p \in V_m} f(p) \left(\frac{1}{a_m} \sum_{q \in V_m} g(q) G_{2s}^m(p, q) \right) \\ &= \frac{1}{a_m} \sum_{p \in V_m} f(p) (-\Delta_m)^{-2s} g(p) \\ &= \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p) \end{aligned}$$

□

Example 3.3: Here we will give an example of a discrete fractional Gaussian field. So Let $(W_i)_{1 \leq i \leq N_m}$ be a sequence of i.i.d Gaussian random variables with mean zero and variance one, i.e. white noise. Then define

$$X_s^m(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) W_i, \quad x \in V_m.$$

First we check the mean, so

$$\mathbb{E}[X_s^m] = \begin{pmatrix} \mathbb{E}[X_s^m(x_1)] \\ \vdots \\ \mathbb{E}[X_s^m(x_{N_m})] \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

since

$$\mathbb{E}[X_s^m(x_i)] = \mathbb{E}\left[\sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) W_i\right] = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \mathbb{E}[W_i] = 0.$$

Moving on to the covariance matrix we have that

$$\begin{aligned} \Sigma &= \mathbb{E}\left((X_s^m - \mathbb{E}[X_s^m])((X_s^m)^T - \mathbb{E}[X_s^m]^T)\right) = \\ &= \mathbb{E}\left[X_s^m \cdot (X_s^m)^T\right] = \mathbb{E}\left[\begin{pmatrix} X_s^m(x_1)X_s^m(x_1) & \cdots & X_s^m(x_1)X_s^m(x_{N_m}) \\ \vdots & \ddots & \vdots \\ X_s^m(x_{N_m})X_s^m(x_1) & \cdots & X_s^m(x_{N_m})X_s^m(x_{N_m}) \end{pmatrix}\right]. \end{aligned}$$

Looking at a single element $\Sigma_{p,q}$ we find

$$\begin{aligned} \mathbb{E}[X_s^m(x_p)X_s^m(x_q)] &= \mathbb{E}\left[\left(\sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x_p) W_i\right) \left(\sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \phi_j^m(x_q) W_j\right)\right] \\ &= \mathbb{E}\left[\sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \phi_i^m(x_p) \phi_i^m(x_q) W_i^2\right] + \mathbb{E}\left[\sum_{i,j=1, i \neq j}^{N_m} (\lambda_i^m \lambda_j^m)^{-s} \phi_i^m(x_p) \phi_j^m(x_q) W_i W_j\right] \\ &= \sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \phi_i^m(x_p) \phi_i^m(x_q) \mathbb{E}[W_i^2] + \sum_{i,j=1, i \neq j}^{N_m} (\lambda_i^m \lambda_j^m)^{-s} \phi_i^m(x_p) \phi_j^m(x_q) \mathbb{E}[W_i] \mathbb{E}[W_j] \\ &= \sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \phi_i^m(x_p) \phi_i^m(x_q) \\ &= G_{2s}^m(x_p, x_q) \end{aligned}$$

using that W_i are independent. Thus X_s^m is a discrete fractional Gaussian field.

3.2 Fractional Gaussian Fields

Now that we have the fractional Laplacian let us state the fractional Gaussian field.

Definition 3.4: Let $s \geq 0$. A *fractional Gaussian field* X_s with parameter s on K is a centered Gaussian field $\{X_s(f) : f \in S(K)\}$ such that for any $f, g \in S(K)$

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu$$

We now quickly move to proving that the fractional Gaussian fields exist. First we will state the Bochner-Minlos theorem for nuclear spaces, since it is needed for the proof of existence. It will be given without proof but one can be found in [Joh03].

Theorem 3.5 (Bochner-Minlos): Assume $\mathcal{X}$ is a nuclear space. Any continuous positive-definite linear functional Λ on $\mathcal{X}$ satisfying $\Lambda(0) = 1$ is the Fourier transform of a countably additive positive normalized measure on $\mathcal{X}'$

We will quickly state and prove a lemma, making it easier to prove a certain functional is positive definite.

Lemma 3.6: The functional $\Phi : \mathcal{X} \rightarrow \mathbb{R}$ defined as $\Phi(v) = \exp(-\frac{1}{2}\langle v, v \rangle)$ is positive definite.

Proof. Let $v_1, \dots, v_n$ be elements of $\mathcal{X}$ and choose an orthonormal basis $e_1, \dots, e_m$ of $\text{span}\{v_1, \dots, v_n\}$. Further let $Z = (Z_1, \dots, Z_m)$ be a vector of independent standard normal real random variables, and note that for all $u \in \mathbb{R}^m$ we have that

$$\Phi\left(\sum_{j=1}^{\infty} u_j e_j\right) = \exp\left(-\frac{1}{2} \sum_{i=1}^m u_i^2\right) = \mathbb{E}\left[e^{iu \cdot Z}\right].$$

How can $u \in \mathbb{R}^n$ be decomposed with another basis. This leads us to for $z \in \mathbb{R}^n$? where is z from? + definition of semi definite

$$z^T A^\Phi z = \sum_{j,k=1}^n z_j \bar{z}_k \Phi(v_j - v_k) = \sum_{j,k=1}^n z_j \bar{z}_k \mathbb{E}\left[e^{i(v_j - v_k) \cdot Z}\right] = \mathbb{E}\left[\left|\sum_{j=1}^n z_j e^{iv_j \cdot Z}\right|^2\right] \geq 0$$

where does the last equality come from? □

Theorem 3.7: Let $s \geq 0$, there exists a centered Gaussian distribution X_s on $S'(K)$ such that for $f, g \in S(K)$

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu$$

Proof. The space $S(K)$ is a nuclear space why it is enough to use theorem 3.5(Bochner-Minlos) on the functional

$$\Lambda : f \rightarrow \exp\left(-\frac{1}{2} \|f\|_k^2\right).$$

This is enough since theorem 3.5 then gives an unique probability measure ν such

$$\exp\left(-\frac{1}{2} \|f\|_k^2\right) = \int_{S'(K)} e^{i\langle x, f \rangle} \, d\nu(x).$$

Notice that the right side is an characteristic function for some random variable. If we choose $f = t \cdot g$ for some $g \in S(K)$ and $t \in \mathbb{R}$ (note that $t \cdot g \in S(K)$) then the left side of the equation becomes the characteristic function for the normal distribution with mean zero and variance $\|f\|_s^2$. Thus we have the X_s we wanted.

Now to prove that Λ satisfies the requirements of theorem 3.5. Obviously $\Lambda(0) = 1$, further it follows from the fact that $\|t \cdot f\|_k^2$ is an inner product that with lemma 3.6 Λ is positive definite. Lastly we need to prove it is continuous at zero.

First we note that since $(-\Delta)^{-s}$ is a bounded operator we have that there exists some constant $C > 0$ so $\|(-\Delta)^{-s} f\|_{L^2(K, \mu)} \leq C \|f\|_{L^2(K, \mu)}$.

Now let $\epsilon > 0$ be given, choose

$$\delta = \begin{cases} \left(\frac{-2 \cdot \ln(1-\epsilon)}{C}\right)^{\frac{1}{2}}, & \text{when } \epsilon \in (0, 1) \\ 1, & \text{when } \epsilon \geq 1. \end{cases}$$

First if $\epsilon \geq 1$ notice that

$$\left| \exp\left(-\frac{1}{2} \|f\|_s^2\right) - \exp\left(-\frac{1}{2} \|0\|_s^2\right) \right| \leq 1 \leq \epsilon, \quad \forall f \in S(K)$$

in which we can choose δ to be any real number larger than 0. Moving on to $\epsilon \in (0, 1)$ we get

$$\begin{aligned}
 \left| \exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right) \right| &\leq \left| \exp\left(-\frac{1}{2}C\|f\|_{L^2(K,\mu)}^2\right) - 1 \right| \\
 &= \left| \exp\left(-\frac{1}{2}C \int_K |f|^2 d\mu\right) - 1 \right| \\
 &\leq \left| \exp\left(-\frac{1}{2}C \int_K \delta^2 d\mu\right) - 1 \right| \\
 &= \left| \exp\left(-\frac{1}{2}C \int_K \frac{-2 \cdot \ln(1-\epsilon)}{C} d\mu\right) - 1 \right| \\
 &= \left| \exp(\ln(1-\epsilon)) - 1 \right| \\
 &= \epsilon
 \end{aligned}$$

in the third to last equality using that μ is a probability measure. In which case we have continuity in 0 as desired. $\square$

In the following we want to prove that a FGF has an L^2 density **What is it?** if and only if $s > \frac{d_h}{2d_w}$. But for this we need to construct white noise.

Let $\mathcal{K}$ be the Borel σ -algebra on K , then we can define

Let W the measure which is uniquely defined by the char. function

$$\int_{S'(K)} \exp(i\langle \omega, \phi \rangle) dW(\omega) = \exp\left(-\frac{1}{2}\|\phi\|_{L^2}^2\right)$$

for $\phi \in S(K)$. This measure again comes from theorem 3.5. **I dunno if this is correct**

This will be done following [Le 16].

This whole section then since μ is a finite measure and therefore σ -finite. It then follows from [Le 16, Prop. 1.13] that there exists a probability field $(\Omega, \mathcal{F}, \mathbb{P})$ with a real-valued centered Gaussian random measure $W : L^2(\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathcal{K}$ with **density/intensity?** μ .

If we now write $W(A) = W$

Definition 3.8: Let $s > \frac{d_h}{2d_w}$. The fractional Brownian field with parameter s is defined as

$$\tilde{X}_s(x) = \int_K G_s(x, y) W(dy)$$

Note that since $s > \frac{d_h}{2d_w}$ we can write

$$\tilde{X}_s(x) = \sum_{i=1}^{\infty} \lambda_i^{-s} \phi_i(x) W_i$$

where $W_i = W(\phi_i)$ is an i.i.d sequence of Gaussian random variables with mean zero and variance one.

We will now show that one can construct a fractional Gaussian Field from the fractional Brownian field

Proposition 3.9: Let $s > \frac{d_h}{2d_w}$. Then the Gaussian random field defined by

$$X_s(f) = \int_K f(x) \tilde{X}_s(x) d\mu(x), \quad f \in S(K)$$

has the law of a fractional Gaussian field with parameter s . Moreover, if there exists a Gaussian field $(\tilde{X}_s(x))_{x \in K}$ on K with

$$\mathbb{E} \left[\int_K \tilde{Y}_S(x)^2 d\mu(x) \right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x) \tilde{Y}_s(x) \, d\mu(x), \quad f \in S(K)$$

has the law of fractional Gaussian field with parameter s , then $s > \frac{d_h}{2d_w}$

Proof. Assume $s > \frac{d_h}{2d_w}$. The from Fubini's theorem we have for every $f \in S(K)$ that a.s

$$\int_K f(x) \tilde{X}_s(x) \, d\mu(x) = \int_K f(x) \int_K G_s(x, y) W(dy) \, d\mu(x) = \int_K (-\Delta)^{-s} f(y) W(dy)$$

in which case by definition of W we have that $X_s(f) = \int_K f(x) \tilde{X}_s(x) \, d\mu(x)$ is a Gaussian random variable with mean zero and variance

$$\int_K |(-\Delta)^{-s} f(x)|^2 \, d\mu(x).$$

and thus a fractional Gaussian field with parameter s .

Now assume that there is a Gaussian field $(\tilde{X}_s(x))_{x \in K}$ on K with

$$\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x) \tilde{Y}_s(x) \, d\mu(x), \quad f \in S(K)$$

has the law of fractional Gaussian field with parameter s . Then by spectral decomposition [reference?](#) we have

$$\tilde{Y}_s(x) = \sum_{i=1}^{\infty} Y_s(\phi_i) \phi_i(x).$$

Note that $(Y_s(\phi_i))_{i \geq 1}$ is a sequence of independent Gaussian random variables with mean zero [why mean zero?](#) and [why independent Gaussian?](#) and variance $(\lambda_i^s)_{i \geq 1}$. By definition 3.4 and proposition 2.27 we have that

$$\mathbb{E} [Y_s(\phi_i) Y_s(\phi_j)] = \int_K (-\Delta)^{-s} \phi_i (-\Delta)^{-s} \phi_j \, d\mu = \frac{1}{(\lambda_j \lambda_i)^s} \int_K \phi_i \phi_j \, d\mu = \frac{1}{(\lambda_j \lambda_i)^s} \delta_{ij}.$$

Since $\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] < \infty$ we have that

$$\begin{aligned} \mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] &= \mathbb{E} \left[\int_K \left(\sum_{i=1}^{\infty} Y_s(\phi_i) \phi_i(x) \right)^2 \, d\mu(x) \right] \\ &= \mathbb{E} \left[\sum_{i=1}^{\infty} Y_s(\phi_i)^2 \int_K \phi_i(x) \phi_i(x) \, d\mu(x) \right] \\ &= \sum_{i=1}^{\infty} \mathbb{E} [Y_s(\phi_i)^2] \\ &= \sum_{i=1}^{\infty} \frac{1}{\lambda_i^{-2s}} < \infty \end{aligned}$$

where we use that $\int_K (\sum_{i=1}^{\infty} Y_s(\phi_i) \phi_i(x))^2 \, d\mu(x) = \int_K \sum_{i=1}^{\infty} Y_s(\phi_i)^2 \phi_i(x)^2 \, d\mu(x)$ which stems from the fact that $\int_K \phi_i \phi_j \, d\mu = \delta_{ij}$ and $Y_s(\phi_i)$ is a constant wrt. $\mu(x)$.

We now have that

$$\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] = \sum_{i=1}^{\infty} \frac{1}{\lambda_i^{-2s}} < \infty$$

in which case by lemma 2.25 we have that $s > \frac{d_h}{2d_w}$ □

Chapter 4

Main Theorem

Now that we have presented and proved the existence of fractional Gaussian fields X_s on K and discrete fractional Gaussian fields X_s^m on V_m our next task is proving they converge. Therefore in this section our main goal is to show that X_s^m converges in distribution to X_s .

4.1 Preliminary lemmas

In this section we will prove a collection of lemmas that will be needed later.

Lemma 4.1: Let $\{P_t\}_{t \geq 0}$ be the semigroup generated by the Dirichlet Laplacian $(-\Delta)^{-s}$ and $p_t(x, y)$ the heat kernel admitted by $\{P_t\}_{t \geq 0}$. Then if ϕ_i is an orthonormal eigenfunction of $(-\Delta)^{-s}$ then

$$P_t \phi_i(x) = e^{-\lambda_i t} \phi_i(x)$$

Proof. It follows immediately from the calculations:

$$\begin{aligned} P_t \phi_i(x) &= \int_K p_t(x, y) \phi_i(y) \, d\mu(y) = \int_K \left(\sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y) \right) \phi_i(y) \, d\mu(y) \\ &= \int_K e^{-\lambda_i t} \phi_i(x) \, d\mu(y) = e^{-\lambda_i t} \phi_i(x) \end{aligned}$$

□

In the following proof we will use results from [Kur69]. Before stating the lemma and proof we will give some arguments as to why we are allowed to use these results.

Firstly we note that since Δ is a bounded operator then by [Paz92, Theorem 1.2] it is the infinitesimal generator of a uniformly continuous semigroup.

Secondly we have that the sequence of **laplacians** $\{\Delta_m\}_{m \geq 0}$ indeed coincides with extended limit defined in [Kur69, page 355]. This follows from theorem 2.16.

Lastly the range $\mathcal{R}(\lambda_0 - \Delta)$ is dense in L^2 **why?**

Lemma 4.2: For all $f \in S(K)$ and $t > 0$,

$$\lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) = \int_K f(x) P_t f(x) \, d\mu(x)$$

where $f_m = f|_{V_m}$.

Proof. In this proof we wish to use [Kur69, theorem 2.1] so recall that by the definition of the Laplacian on K we have that

$$\lim_{m \rightarrow \infty} \sup_{p \in V_m} |\Delta_m f_m(p) - \Delta f(p)| = 0.$$

Thus by [Kur69, theorem 2.1] **Is defined with semigroup $T(s)$ and page 354, is P_n in our case $\int_K \psi_x^{(m)} \, d\mu$** we have that for all $t \geq 0$

$$\lim_{m \rightarrow \infty} \sup_{p \in V_m} |P_t^m f_m(p) - P_t f(p)| = 0. \quad (4.1)$$

So if we now write

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) = \int_{V_m} f_m P_t^m f_m \, d\mu_m.$$

Thus

$$\int_{V_m} f_m P_t^m f_m \, d\mu_m = \int_{V_m} f_m (P_t^m f_m - (P_t f)_m) \, d\mu_m + \int_{V_m} f_m (P_t f)_m \, d\mu_m.$$

The first integral converges to 0 as $m \rightarrow \infty$ because of (4.1). Further by lemma we have $\int_{V_m} f_m (P_t f)_m \, d\mu_m \rightarrow \int_K f P_t f \, d\mu$ as $m \rightarrow \infty$. □

Lemma 4.3: For all $f \in S(K)$ and $s \geq 0$ the following holds

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) \rightarrow \int_K f(x) (-\Delta)^{-2s} f(x) \, d\mu(x), \quad \text{as } m \rightarrow \infty$$

Proof. For $s = 0$ we find that

$$\begin{aligned} (-\Delta_m)^{-0} f &= \sum_{i=1}^{N_m} \frac{1}{(\lambda_i^m)^0} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \psi_i^m(y) f(y) \\ &= \sum_{i=1}^{N_m} \phi_i^m(x) \int_{V_m} \phi_i^m(y) f(y) \, d\mu_m \\ &= f_m. \end{aligned}$$

Likewise by definition of $(-\Delta)^{-s}$ we find

$$(-\Delta)^{-0} f = \sum_{i=1}^{\infty} \phi_i \int_K \phi_i(y) f(y) \, d\mu = f.$$

and the lemma follows from corollary 1.15. Now assume $s > 0$ then by Fubini's theorem and the fact that the semi-group is linear we have that

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) (-\Delta_m)^{-2s} f_m(p) = \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) \, dt.$$

Now by lemma 4.1 and the definition of f_m we find that

$$\begin{aligned} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m \sum_{j=1}^{\infty} \langle f_m, \phi_j \rangle \phi_j \\ &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^{\infty} \langle f_m, \phi_j \rangle P_t^m \phi_j \\ &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^{\infty} \langle f_m, \phi_j \rangle e^{-\lambda_j^m t} \phi_j \\ &\leq e^{-\lambda_1^m t} \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 \end{aligned}$$

where we can guarantee the inequality for all j since $0 < \lambda_1 < \lambda_2 < \lambda_3 < \dots$

Note that $\sup_m \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 < \infty$ since $f \in S(K) \subset C(K)$ so f is bounded and $\#V_m < \infty$ for all m . We can now by dominated convergence, Fubini's theorem and lemma 4.2 find that

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) &= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} t^{2s-1} \lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) \, dt \\ &= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \int_K f(x) P_t f(x) \, d\mu(x) \, dt \\ &= \frac{1}{\Gamma(2s)} \int_K f(x) \int_0^\infty t^{2s-1} P_t f(x) \, dt \, d\mu(x) \\ &= \int_K f(x) (-\Delta)^{-2s} f(x) \, d\mu(x) \end{aligned}$$

as desired. $\square$

4.2 Main Theorem

With everything in mind we are now ready to prove the convergence of discrete fractional Gaussian fields to fractional Gaussian fields.

Theorem 4.4: Let $s \geq 0$, then X_s^m converges to X_s in distribution in $S'(K)$ when $m \rightarrow \infty$

Proof. First since $S(K)$ is a nuclear space, by [BDW17, Corollary 2.4] it is enough to prove the convergence of the characteristic functional, e.i that for all $f \in S(K)$

$$\mathbb{E} [\exp(itX_s^m(f_m))] \rightarrow \mathbb{E} [\exp(itX_s(f))], \quad \text{as } m \rightarrow \infty$$

where $f_m = f|_{V_m}$. It follows from proposition 3.2 that

$$\mathbb{E} [\exp(itX_s^m(f_m))] = \exp \left(-\frac{1}{2} t^2 \mathbb{E} [(X_s^m(f_m))^2] \right) = \exp \left(-\frac{1}{2a_m} t^2 \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) \right)$$

Likewise we find that via definition 3.4 we get that

$$\mathbb{E} [\exp(itX_s(f))] = \exp \left(-\frac{1}{2} t^2 \mathbb{E} [(X_s(f))^2] \right) = \exp \left(-\frac{1}{2} t^2 \int_K f(-\Delta)^{-2s} f \, d\mu \right).$$

Where we notice that from corollary 2.34

$$\begin{aligned} \mathbb{E} [(X_s(f))^2] &= \int_K (-\Delta)^s f (-\Delta)^s f \, d\mu \\ &= \int_K \int_K f(p) f(q) G_{2s}(p, q) \, d\mu(p) \, d\mu(q) \\ &= \int_K f(-\Delta)^{-2s} f \, d\mu. \end{aligned}$$

The proof now follows from lemma 4.3. $\square$

4.3 Simulations

As a part of this thesis we have created and simulated a discrete fractional Gaussian field on the Vicsek set V_m . The precise field we have constructed was the one given in example 3.3. These simulation have had a varying rate of success, as we have been able to simulate $X_s^m(x) = \sum \lambda_j \phi_j$ for a reasonable set of s -values, we have hit a bottleneck in the fact that we are not able to simulate X_s^m for $m \geq 7$. The reason for this bottleneck lies in remark 2.2.

As stated in the remark for any function $f : V_m \rightarrow \mathbb{R}$ we can define a matrix L such that

$$L_m F = \begin{pmatrix} \delta_m f(x_1) \\ \vdots \\ \Delta_m f(x_{N_m}) \end{pmatrix}, \quad \text{where } F = \begin{pmatrix} f(x_1) \\ \vdots \\ f(x_{N_m}) \end{pmatrix}.$$

It is precisely from this matrix L_m we compute the eigenvalues and eigenfunctions of $-\Delta_m$. One shall now note that the amount of points in V_m is $5^m \cdot 4 + 1$ so as m grows this number explodes (see ??), and there is not any smart way to construct the matrix L_m ¹. Thus the method of constructing the matrix L_m was simply by constructing the Vicsek set V_m and manually checking each point for neighbours.

m	$\#V_m$
1	21
2	101
3	501
4	2501
5	12501
6	62501
7	312501
10	39062501
20	$3.81 \cdot 10^{14}$

Table 4.1: Points in V_m

Construction of the Vicsek set for simulation

While the construction of the Vicsek set is not terrible, it is also not good. The way it was done is by first generating the first "plus" V_0 . Then going from V_0 to V_1 we find the endpoints (Tips of the plus) of V_0 and then just add V_0 to it.

```

1 def findPointsForLineCollection(self, endPoints: list = None, points: list =
    None, n: int = None, init_length: int = None) -> list:
2
3
4     # Setup if none is given as argument
5     if n == None:
6         n = self.n
7     if init_length == None:
8         init_length = 3
9     if points == None:
10        points = [[np.array([-init_length, 0]), np.array([init_length, 0])], [np.array
            ([0, init_length]), np.array([0, -init_length])]]
11
12    # This is to check if the recursion given is smaller than zero
13    if n <= 0:
14        # This checks if we are in the first iteration or if we have just reached
            recursion depth zero
15    if ([[np.array([-init_length, 0]), np.array([init_length, 0])]] and [[np.
            array([0, init_length]), np.array([0, -init_length])]]) in points):
16        return []
17    else:
18        return [[np.array([-init_length, 0]), np.array([init_length, 0])], [np.array
            ([0, init_length]), np.array([0, -init_length])]]
19    else:
20        collection = []
21

```

¹That we have found while researching [Formulation of this?](#)

```
22 # Setup if none is given as argument
23 if endPoints == None:
24     endPoints = [np.array([-init_length, 0]), np.array([init_length, 0]),
25                 np.array([0, init_length]), np.array([0, -init_length])]
26
27 # Finds all the points in the current iteration
28 for x in endPoints:
29     collection.extend([[y + x*2 for y in lists] for lists in points])
30
31 # Find the new endpoints
32 newEndPoints = [x + 2*x for x in endPoints]
33
34 return collection + self.findPointsForLineCollection(newEndPoints, collection
    + points, n-1, init_length)
```

1. The rækkefølge af $x_i \in V_m$ er den samme hver gang på grund af sets.

I have some pretty pictures. Want to see the fields $X_s^m(x) = \sum \lambda_j \phi_j$ to $\tilde{X}_s$?

Bibliography

- [Bau25] Fabrice Baudoin. *A short introduction to Dirichlet Spaces*. 2025.
- [BC22] Fabrice Baudoin and Li Chen. “Dirichlet fractional Gaussian fields on the Sierpinski gasket and their discrete graph approximations”. In: (2022). eprint: [arXiv:2201.03970](https://arxiv.org/abs/2201.03970).
- [BDW17] Hermine Biermé, Olivier Durieu, and Yizao Wang. *Generalized random fields and Lévy’s continuity theorem on the space of tempered distributions*. 2017. arXiv: 1706.09326 [math.PR]. URL: <https://arxiv.org/abs/1706.09326>.
- [Chu06] Fan R. K. Chung. *Spectral Graph Theory*. 2006. URL: <https://mathweb.ucsd.edu/~fan/research/cb/ch1old.pdf>.
- [Dod81] Jozef Dodziuk. “Eigenvalues of the Laplacian and the Heat Equation”. In: *The American Mathematical Monthly* 88.9 (1981), pp. 686–695. ISSN: 00029890, 19300972. URL: <http://www.jstor.org/stable/2320674> (visited on 05/22/2025).
- [Fal03] Kenneth John Falconer. *Fractal Geometry - Mathematical Foundations and Applications*. English. 2nd. Second Edition (major revision plus solutions manual). John Wiley & Sons, Ltd, 2003. ISBN: 978-0-470-84862-3.
- [FS92] M. Fukushima and T. Shima. “On a spectral analysis for the Sierpinski gasket”. In: (1992). DOI: <https://doi.org/10.1007/BF00249784>.
- [Ger23] Alberto Gervani. “Hausdorff dimension and Iterated Function Systems”. Bachelor Thesis. Dipartimento di Matematica “Tullio Levi-Civita”, Università degli Studi di Padova, Sept. 2023.
- [Haz22] Mona Hazeem. *Similarities and Hausdor dimension in fractal geometry*. Självständiga arbeten i matematik. Matematiska Institutionen, Stockholms Universitet, Jan. 2022.
- [Hen99] Jeffrey T. Henrikson. *Completeness and Total Boundedness of the Hausdorff Metric*. 1999. URL: <https://web.archive.org/web/20020623095720/http://www-math.mit.edu/phase2/UJM/vol1/HAUSF.PDF>.
- [Joh03] Troels Roussau Johansen. “The Bochner – Minlos Theorem for nuclear spaces and an abstract white noise space”. In: (2003). URL: <https://web.archive.org/web/20160304081333/http://www3.math.uni-paderborn.de/~johansen/seminars/minlos.pdf>.
- [Kig01] Jun Kigami. *Analysis on Fractals*. Cambridge Tracts in Mathematics. Cambridge University Press, 2001.
- [Kur69] Thomas G Kurtz. “Extensions of Trotter’s operator semigroup approximation theorems”. In: *Journal of Functional Analysis* 3.3 (1969), pp. 354–375. ISSN: 0022-1236. DOI: [https://doi.org/10.1016/0022-1236\(69\)90031-7](https://doi.org/10.1016/0022-1236(69)90031-7). URL: <https://www.sciencedirect.com/science/article/pii/0022123669900317>.
- [Le 16] Jean-François Le Gall. *Brownian Motion, Martingales, and Stochastic Calculus*. Vol. 274. Jan. 2016. ISBN: 978-3-319-31088-6. DOI: 10.1007/978-3-319-31089-3.
- [Mar98] David Nualart Martin T. Barlow. *Lectures on Probability Theory and Statistics*. Ecole d’Ete de Probabilites de Saint-Flour XXV - 1995. Springer Berlin, Heidelberg, 1998. ISBN: 978-3-540-64620-4.
- [Paz92] Amnon Pazy. *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer New York, NY, 1992. ISBN: 978-0-387-90845-8.

- [Pie66] Albrecht Pietsch. *Über die Erzeugung von (F) -Räumen durch selbstadjungierte Operatoren*. German. Springer, 1966.
- [Ski23] Erik Skibsted. *Notes for a course in advanced analysis*. 2023.
- [Str06] Robert S. Strichartz. *Differential Equations on Fractals: A Tutorial*. Princeton University Press, 2006.
- [Tho22] Steen Thorbjørnsen. *Forelæsningsnoter i Vidergående Sandsynlighedsteori*. 2022.
- [Tho24] Steen Thorbjørnsen. *Introduction to Measure and Integration Theory*. 2024.
- [Trè67] François Trèves. *Topological Vector Spaces, Distributions and Kernels*. Academic Press, 1967.